

PROYECTO: “ECO PARADISE”



LABORATORIO DE MEDICIONES Y ENSAYOS III

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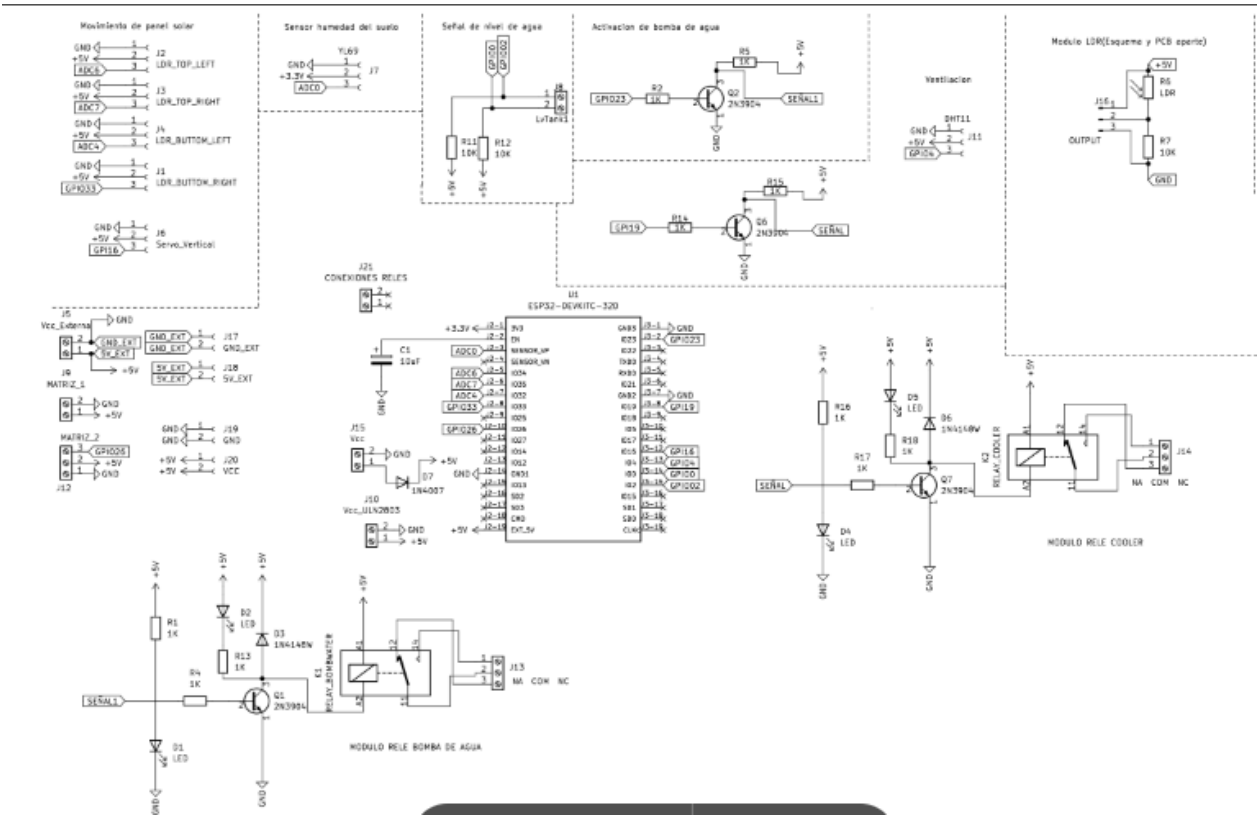
CICLO LECTIVO 2024

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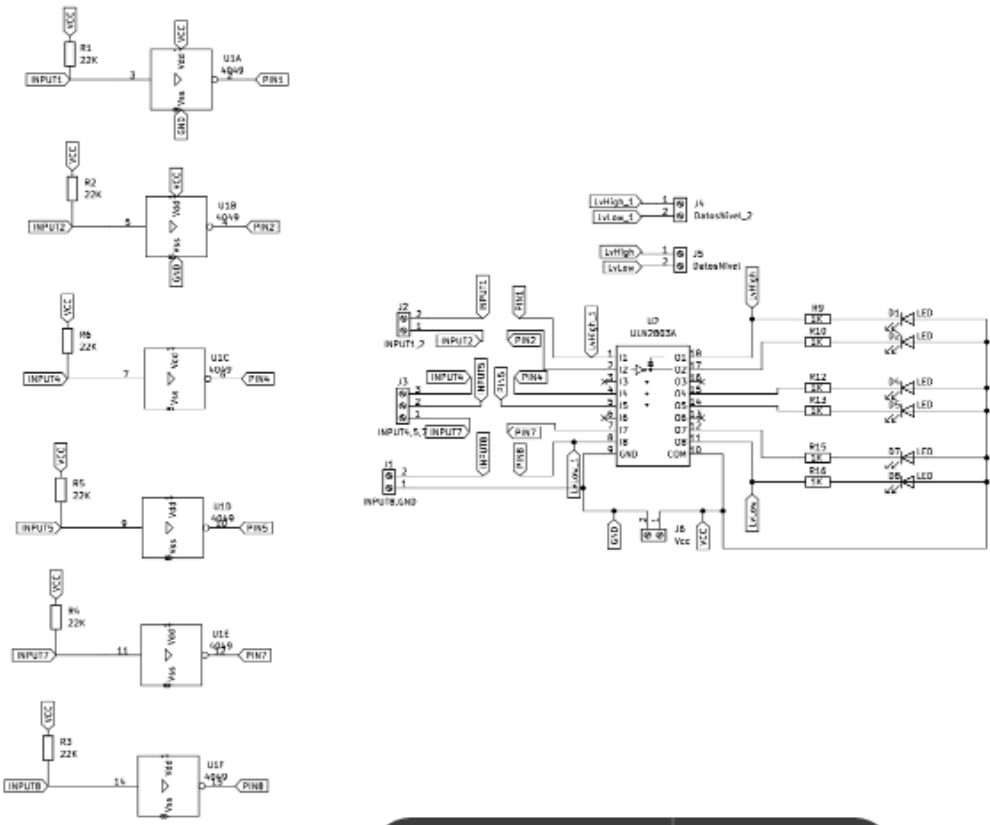
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ESQUEMA ELÉCTRICO

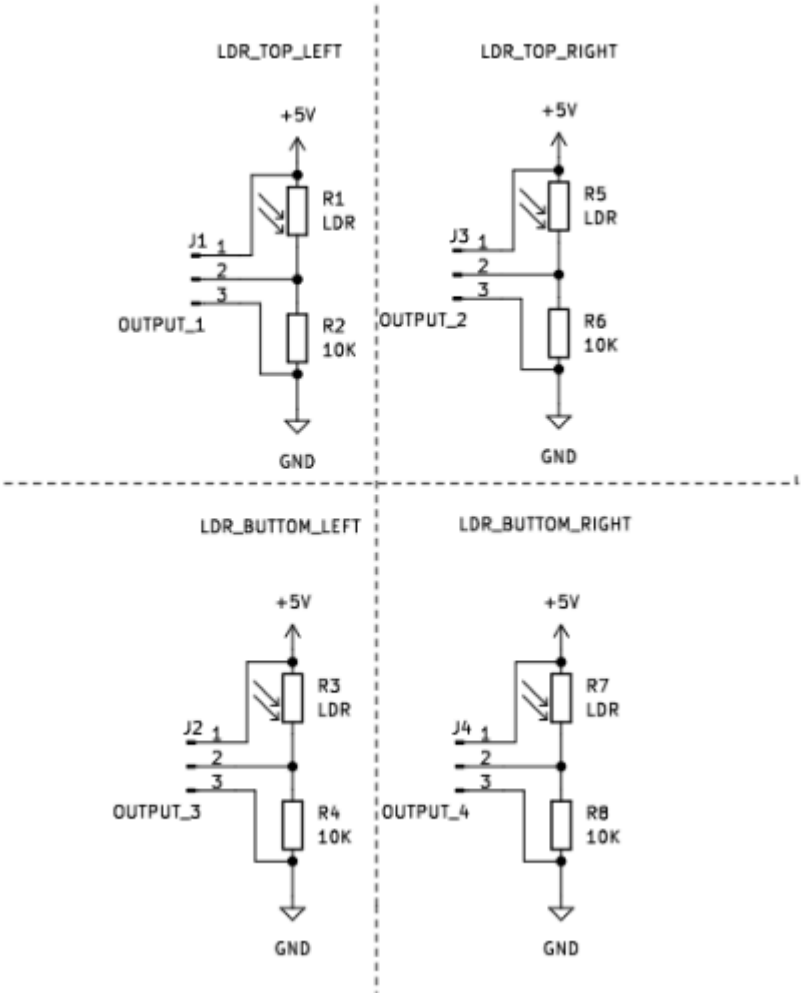
Placa principal:



Placa sensor nivel de agua:



Placa de módulos LDR:



LISTADO ANALÍTICO

Placa Principal

5v-j5(1) 5v-j9(1) 5v-Matriz2(2) 5v-j20(1) 5v-j20(2) 5v-j2(2) 5v-j2(2) 5v-j3(2)
 5v-j1(2) 5v-j6(2) 5v-R11 5v-R12 5v-R5 5v-R15 5v-j11 5v-R6 5v-R16 5v-D5
 5v-D6 5v-Relayk2(a1) 5v-Esp32(19) 5v-D7 5v-j10 5v-R1 5v-D2 5v-D3
 5v-Relayk1(a1) 3,3v-j7(2) 3,3v-Esp32(1) GND-j2(1) GND-j3(1) GND-j4(1)
 GND-j1(1) GND-j6(1) GND-j7(1) GND-Q2(1) GND-Q6(1) GND-j11(1) GND-d4
 GND-Q7 GND-Esp32(14) GND-Esp32(20) GND-Esp32(26) GND-C1(-)
 GND-j15(2) GND-j10(2) GND-j5(2) GND-j9(2) GND-j19(1) GND-j19(2)
 GND-j12(1) GND-D1 GND-Q1(1) ADC6-j2(3) ADC7-j3(3) ADC4-j4(3)
 GPIO33-j1(3) GPI66-j6(3) ADC0-j7(3) R11-J8(1) R11-GPIO02 R12-j8(2)
 R12-GPIO0 GPIO23-R2 R2-Q2(2) Q2(3)--R5 Señal1-Q2(3).R5 GPI19-R14
 R14-Q6(2) Q6(3)--R15 Señal-Q6(3).R15 GPIO4-j11(3) Señal-R16.D4.R17
 R17-Q7(2) Q7(3)--R18.D6.Relayk2(A2) R18-D5 Relayk2(12)--j14(3)
 Relayk2(14)--j14(1) Relayk2(11)--j14(2) Esp32(3)--ADC0 Esp32(5)--ADC6
 Esp32(6)--ADC7 Esp32(7)--ADC4 Esp32(8)--GPIO33 Esp32(10)--GPIO26
 Esp32(21)--GPIO23 Esp32(27)--GPI19 Esp32(31)-GPI16 Esp32(32)--GPIO4
 Esp32(33)--GPIO0 Esp32(34)--GPIO02 Esp32(2)--C1(+) D7-j15(1)
 GND_EXT-J15(2) GND_EXT-J17(1) GND_EXT-j17(2) J5(1)--5v_EXT
 5v_EXT-j18(1) 5v_EXT-j18(2) GPIO26-j12(3) Señal1-R1.R4.D1 R1-Q1(2)
 Q1(3)--R13.D3.RelayK1(A2) R13-D2 Relayk1(12)--j13(3) Relayk1(14)--j13(1)
 Relayk1(11)--j13(2) j21(1)--x j21(2)--x

Nivel Del Agua

VCC–R1 VCC–R2 VCC–R6 VCC–R5 VCC–R4 VCC–R3
VCC–j6.U2(10).D8.D7.D5.D4.D2.D1 VCC–U1A(vdd) VCC–U1B(vdd)
GND–U1A(vss) GND–U1B(vss) GND–j1(1).U2(9).j6(2) INPUT1–R1.U1A(3)
U1A(2)--PIN1 INPUT2–R2.U1B(5) U1B(4)--PIN2 INPUT4–R6.U1C(7)
U1C(6)--PIN4 INPUT5–R5.U1D(9) U1D(10)--PIN5 INPUT7–R4.U1E(12)
U1E(12)--PIN7 INPUT8–R3.U1F(14) U1F(15)--PIN8 D1–R9 R9–LvHigh.U2(18)
D2–R10 R10–U2(17) D4–R12 R12–U2(15) D5–R13 R13–U2(14) D7–R15
R15–U2(12) D8–R16 R16–U2(11).LvLow LvHigh–j5(1) LvLow–J5(2)
LvHigh_1–j4(1) LvLow_1–j4(2) j1(2)--INPUT8 PIN8–U2(8).LvLow_1 PIN7–U2(7)
PIN5–U2(5) PIN4–U2(4) PIN2–U2(2) PIN1–U2(1).LvHigh_1 INPUT7–j3(1)
INPUT5–j3(2) INPUT4–j4(3) INPUT2–j2(1) INPUT1–j2(2)

Módulos Ldr

5v–R1 5v–R5 5v–R3 5v–R7 GND–R2 GND–R6 GND–R4 GND–R8 j1(1)--5v
j2(1)--5v j3(1)--5v j4(1)--5v j1(2)--R1.R2 j2(2)--R3.R4 j3(2)--R5.R6 j4(2)--R7.R8
j1(3)--GND j2(3)--GND j3(3)--GND j4(4)--GND

CÁLCULO DE SUPERFICIE

PLACA PRINCIPAL:

Componentes con forma rectangular:

Cantidad	Componente	$b \cdot h$	$(b \cdot h) \cdot cantidad$	Valor final
2	Modulo rele 5DC	$18mm \cdot 14mm$	$(252\ mm) \cdot 2$	504mm
2	Resistencia SMD 1206 10K Ω	$3,2mm \cdot 1,6mm$	$(5,12\ mm) \cdot 2$	10,24mm
10	Resistencia SMD 1206 1K Ω	$3,2mm \cdot 1,6mm$	$(5,12\ mm) \cdot 10$	51,2mm
1	Diodo 1N4007 THT	$2,36\ mm \cdot 5\ mm$	-	11,8mm
6	Bornera doble	$9,96\ mm \cdot 7,38mm$	$(73,50mm) \cdot 6$	441mm
3	Bornera triple	$15mm \cdot 7mm$	$(105mm) \cdot 3$	315mm
4	2N3904	$3mm \cdot 5mm$	$(15mm) \cdot 4$	60mm
7	Tira de PINs macho 1 x 3 2,54mm	$7mm \cdot 2\ mm$	$(14mm) \cdot 7$	98mm
2	Tira de PINs macho 1x 2 2,54mm	$5mm \cdot 2mm$	$(10mm) \cdot 2$	20mm
2	Tira de PINs hembra 1 x 19 2,54mm	$51mm \cdot 2mm$	$(102mm) \cdot 2$	204mm

Componentes con forma circular:

Cant.	Componente	$r = d \div 2$	$\pi \cdot r^2$	Cant.	Valor final
1	Capacitor electrolítico 100 μ F 16V THT	$r = 5mm \div 2$ $r = 2,5 \text{ mm}$	$\pi \cdot 2,5^2$	-	19,635mm
2	LED 5mm VERDE DIFUSO	$r = 4,95 \text{ mm} \div 2$ $r = 2,475 \text{ mm}$	$\pi \cdot 2,475^2$	19,24 mm	38,49mm
2	LED 5mm ROJO DIFUSO	$r = 4,95 \text{ mm} \div 2$ $r = 2,475 \text{ mm}$	$\pi \cdot 2,475^2$	19,24 mm	38,49mm

Sumatoria de áreas=1811,855 mm

$$\Sigma = 1811,855 \text{ mm} \cdot 6 = 10871,13 \text{ mm}$$

$$b \cdot h = 10871,13 \text{ mm}$$

$$120 \text{ mm} \cdot h = 10871,13 \text{ mm}$$

$$h = 10871,13 \text{ mm} / 120 \text{ mm}$$

$$h = 90,59 \text{ mm}$$

Tamaño de la plaqueta por cálculo de superficie: 120 mm x 90,59 mm

Tamaño final de la plaqueta al momento de hacer el PCB: 120 mm x 90 mm

PLACA SENSOR NIVEL DE AGUA:

Componentes con forma rectangular:

Cantidad	Componente	$b \cdot h$	$(b \cdot h) \cdot \text{cantidad}$	Valor final
6	Resistencia SMD 1206 22K Ω	$3,2 \text{ mm} \cdot 1,6 \text{ mm}$	$(5,12 \text{ mm}) \cdot 6$	30,72 mm
6	Resistencia SMD 1206 1K Ω	$3,2 \text{ mm} \cdot 1,6 \text{ mm}$	$(5,12 \text{ mm}) \cdot 6$	30,72 mm
1	ZÓCALO DIL 16 IC4049	$20 \text{ mm} \cdot 10 \text{ mm}$	—	200mm
1	ULN2803 SMD SOL-18	$11 \text{ mm} \cdot 7 \text{ mm}$	—	77mm
1	Bornera triple	$15 \text{ mm} \cdot 7 \text{ mm}$	105mm	105mm
5	Bornera doble	$9,96 \text{ mm} \cdot 7,38 \text{ mm}$	$(73,50 \text{ mm}) \cdot 5$	367,5 mm

Componentes con forma circular:

Cantidad	Componente	$r = d \div 2$	r	$\pi \cdot r^2$	cantidad	Valor final
2	LED 5mm VERDE DIFUSO	$r = 4,95 \text{ mm} \div 2$	2,475 mm	$\pi \cdot 2,475^2$	19,24·2	38,49mm
2	LED 5mm ROJO DIFUSO	$r = 4,95 \text{ mm} \div 2$	2,475 mm	$\pi \cdot 2,475^2$	19,24·2	38,49mm
2	LED 5mm AMARILLO DIFUSO	$r = 4,95 \text{ mm} \div 2$	2,475 mm	$\pi \cdot 2,475^2$	19,24·2	38,49mm

Sumatoria de áreas=926,41 mm

$$\Sigma = 926,41 \text{ mm} \cdot 5 = 4632,05 \text{ mm}$$

$$b \cdot h = 4632,05 \text{ mm}$$

$$77,5 \text{ mm} \cdot h = 4632,05 \text{ mm}$$

$$h = 4632,05 \text{ mm} / 77,5 \text{ mm}$$

$$h = 59,77 \text{ mm}$$

Tamaño de la plaqueta por cálculo de superficie: 77,5 mm x 59,77 mm

Tamaño final de la plaqueta al momento de hacer el PCB: 77,5 mm x 73,5 mm

PLACA MODULOS LDR:

Componentes con forma rectangular:

Cantidad	Componente	$b \cdot h$	$(b \cdot h) \cdot \text{cantidad}$	Valor final
1	Resistencia SMD 1206 10KΩ	3,2mm·1,6mm	-	5,12 mm
1	Tira de PINs macho 1 x 3 2,54mm 90°	7mm·2 mm	-	14mm

Componentes con forma circular:

Cantidad	Componente	$r = d \div 2$	r	$\pi \cdot r^2$	Valor final
1	LDR 10mm 250mW 300V 600nm	$r = 11,5 \text{ mm} \div 2$	5,75 mm	$\pi \cdot 5,75^2$	103,87mm

Sumatoria de áreas=122,99 mm

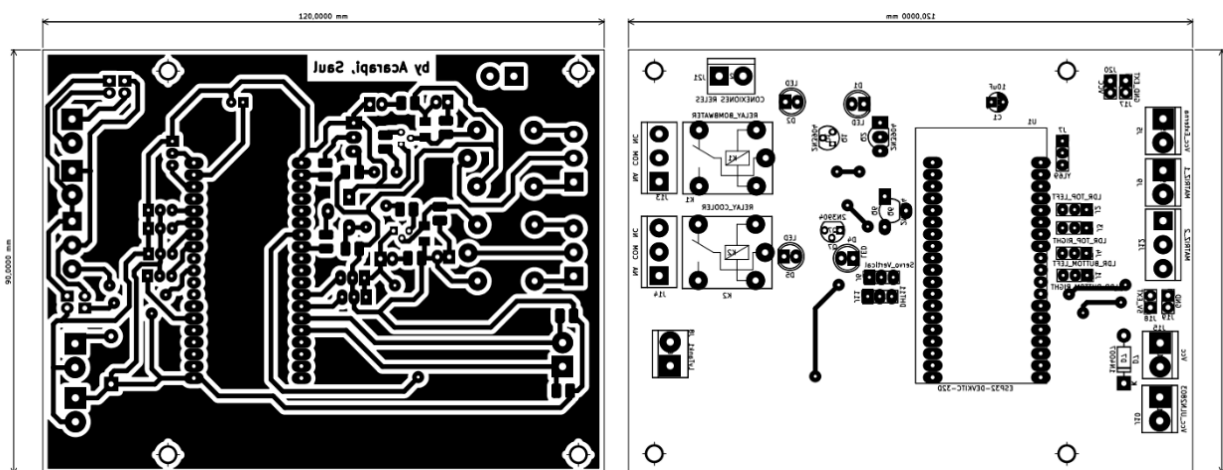
$$\Sigma = 122,99 \text{ mm} \cdot 5 = \sqrt{614,95 \text{ mm}} = 24,8 \text{ mm}$$

Tamaño de la plaqueta por cálculo de superficie: 81,43 mm x 81,43mm

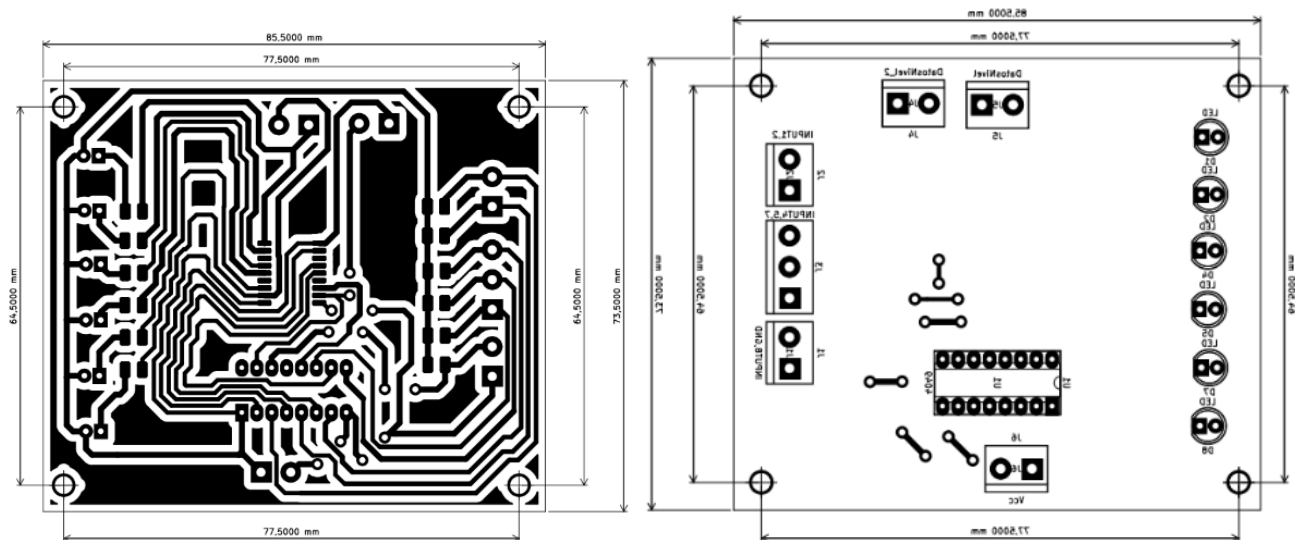
Tamaño final de la plaqueta al momento de hacer el PCB: 71mm x 71mm

DISEÑO PCB (LADO COBRE – LADO COMPONENTES)

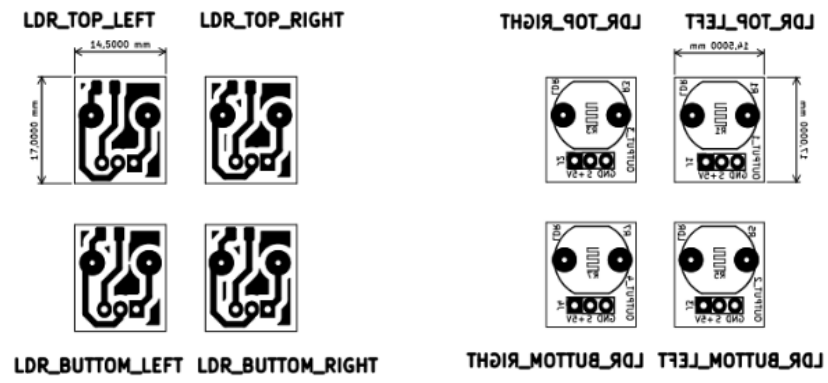
Placa principal(B.Cu y F.Cu):



Placa sensor nivel de agua (B.Cu y F.Cu):

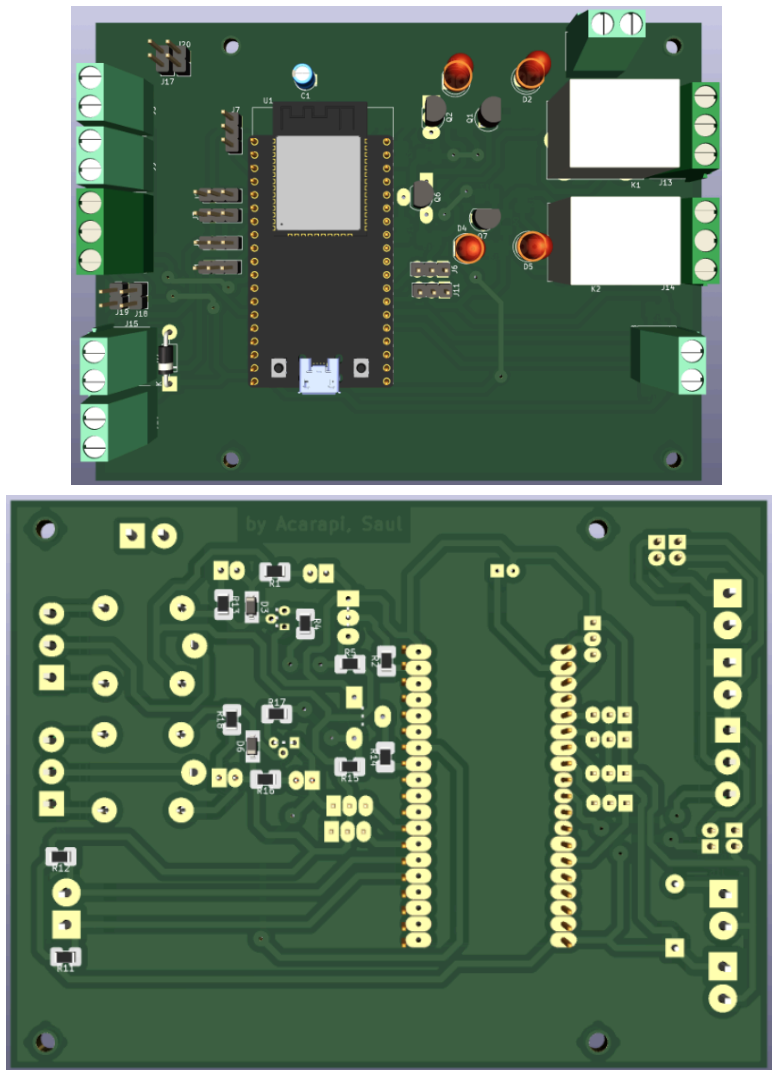


Placa modulos LDR (B.Cu y F.Cu):

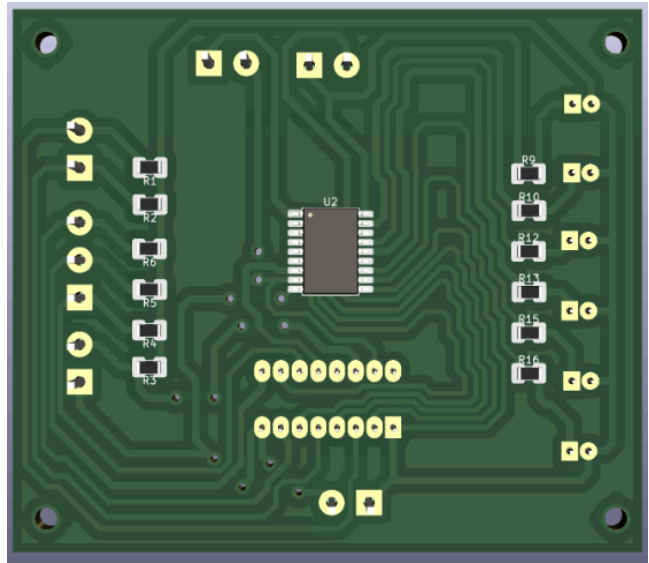
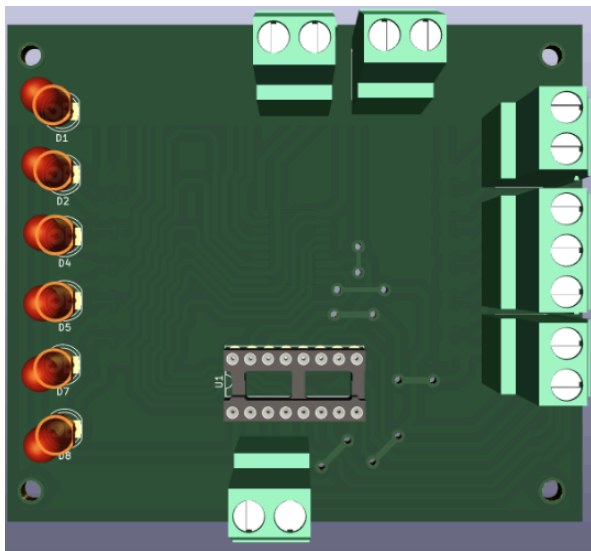


VISTA 3D

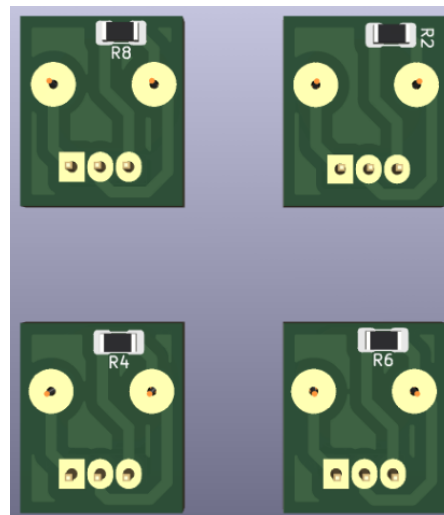
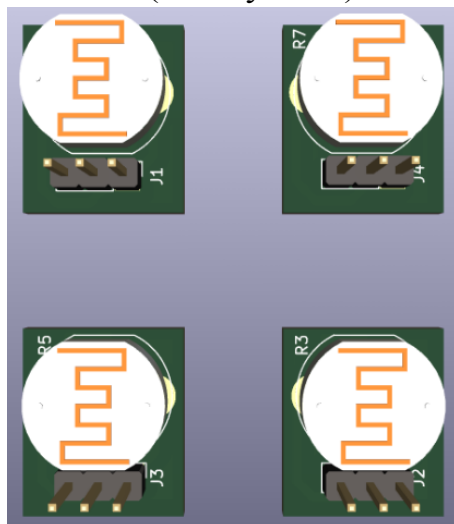
Placa principal (B.Cu y F.Cu):



Placa sensor nivel de agua (B.Cu y F.Cu):



Placa modulos LDR (B.Cu y F.Cu):



PRINCIPIO DE FUNCIONAMIENTO

PLACA PRINCIPAL:

El objetivo de esta placa es conectar todos los componentes, sensores y permitir una comunicación entre estos. Esta usa como unidad de procesamiento el Esp32, Maneja principalmente las luces led de crecimiento, módulos relé del cooler, modulo rele de bomba de agua, sensor de humedad, sensor del nivel del agua y la activación de la bomba de agua y diversos procesos que están en la programación.

PLACA SENSOR NIVEL DE AGUA:

El objetivo de esta placa es detectar el nivel del agua del tanque para activar la bomba de agua dependiendo de la humedad del suelo. Esta placa contiene 2 integrado, uno es el CD4049, que es un circuito integrado CMOS que contiene seis inversores (buffers) con entradas y salidas no invertidas y su función es la de invertir el estado de la señal de bajo a alto provenientes de los tornillos que están en el tanque y evitar la electrolisis en el agua, y el segundo es el ULN2803 que es array de transistores Darlington que se utiliza para amplificar y manejar señales de baja corriente provenientes de los tornillos y mostrar el nivel mediante el encendido de los LEDs.

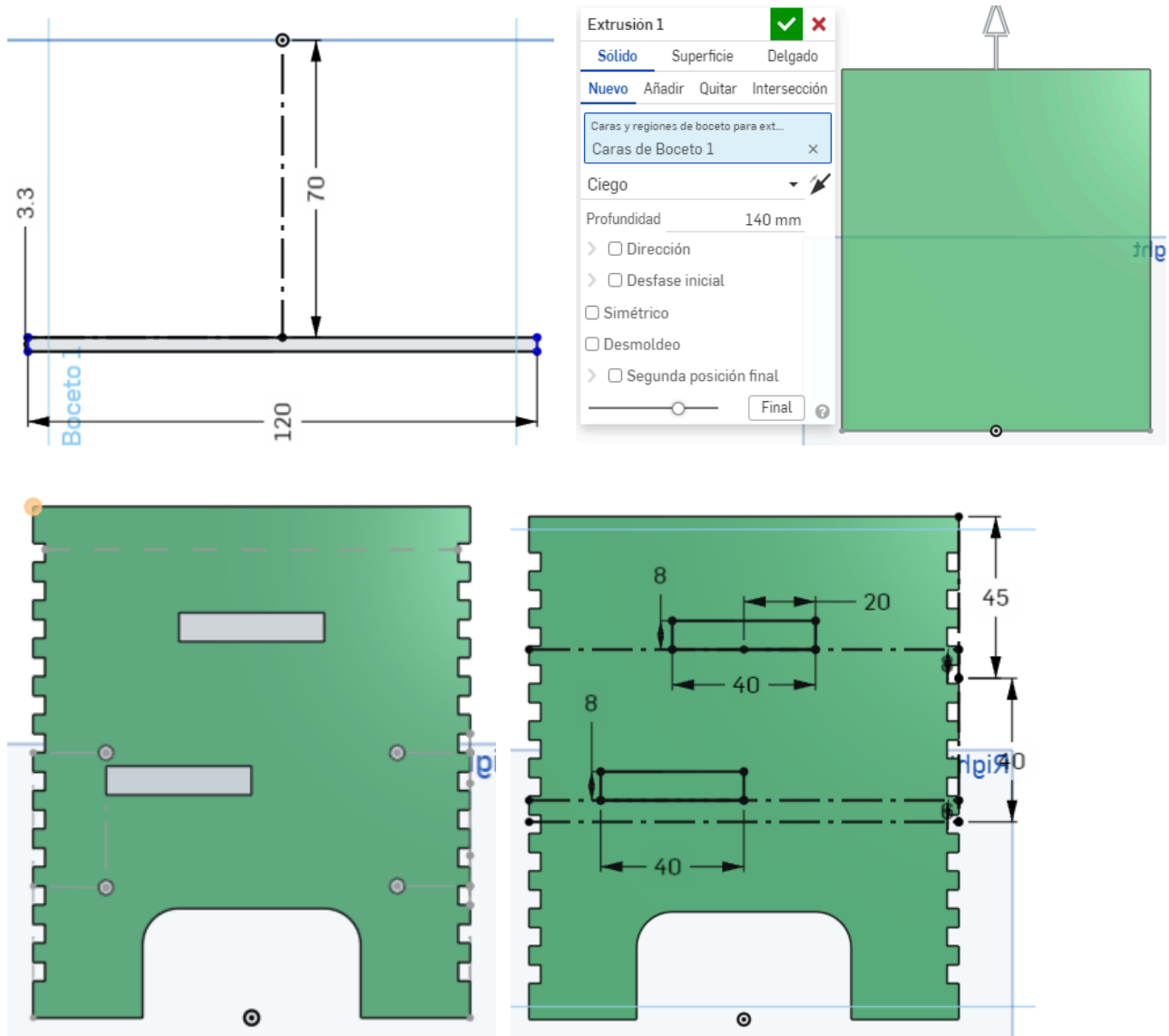
Cuando el agua llega al nivel de un tornillo, este manda la señal al CD4049 el cual invierte la señal de estado bajo a alto y lo manda a una de las entradas del ULN2803, y el transistor correspondiente se activa permitiendo que la corriente fluya a través de la salida, encendiendo el LED conectado a esa salida. La señal del nivel bajo y alto del agua para que el ESP32 pueda saber cuándo activar la bomba es enviando de las salidas, donde se encuentran las LEDs

El ULN2803 es un array de transistores Darlington que se utiliza para amplificar y manejar señales de baja corriente provenientes de los sensores de nivel de agua, y controlar dispositivos de mayor corriente, como los LEDs indicadores.

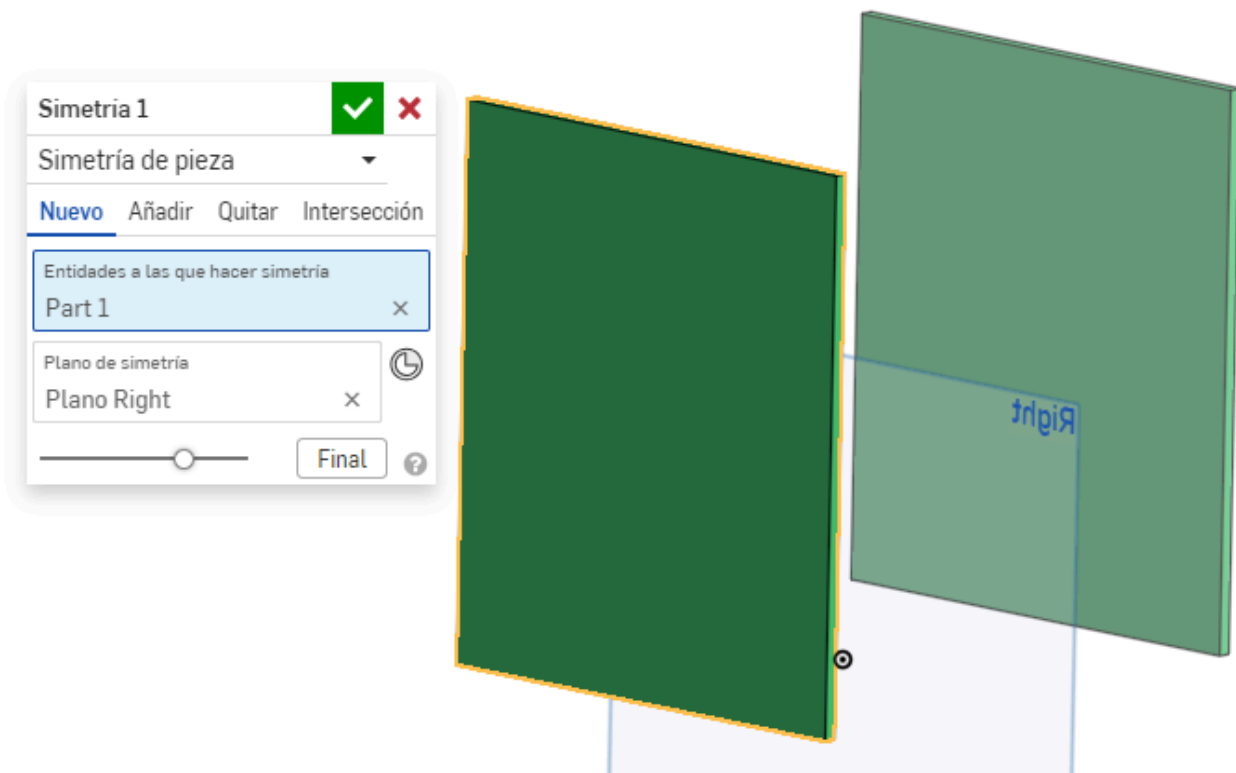
ESQUEMA DE MECANIZADO DE GABINETE

Para el gabinete, se optó por hacerlo de madera (ya que es un prototipo) usando la cortadora láser para cortar la misma. Para hacer el diseño se usó el software online de On Shape y el diseño que se ideó fue el siguiente:

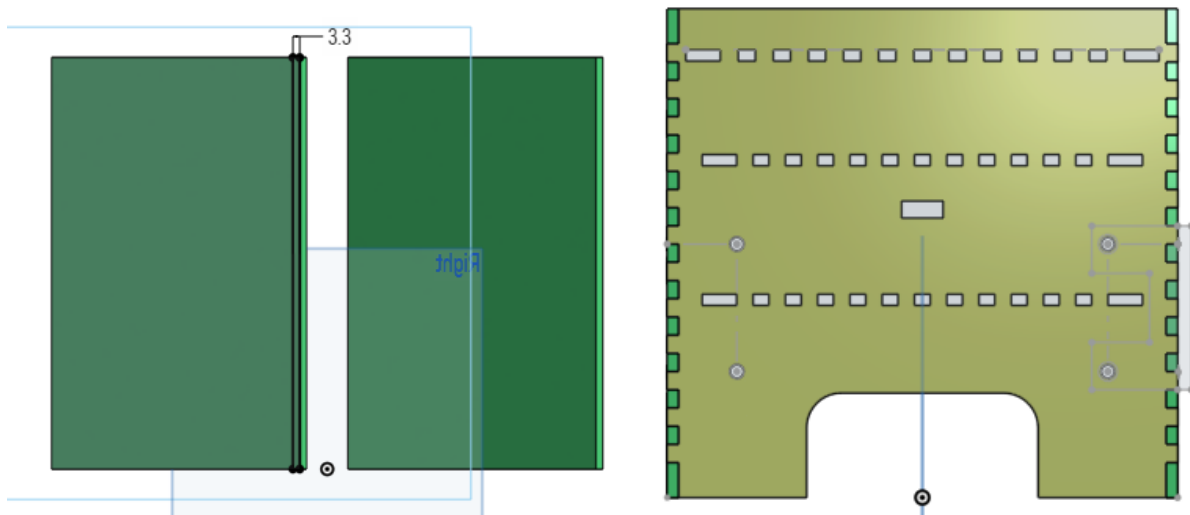
GABINETE PARTE LATERAL IZQUIERDO: En esta pieza se hicieron 2 agujeros rectangulares, el de arriba es para conectar los cables de las señales del tanque del agua y el de abajo es para conectar las matrices RGB con su respectiva alimentación.

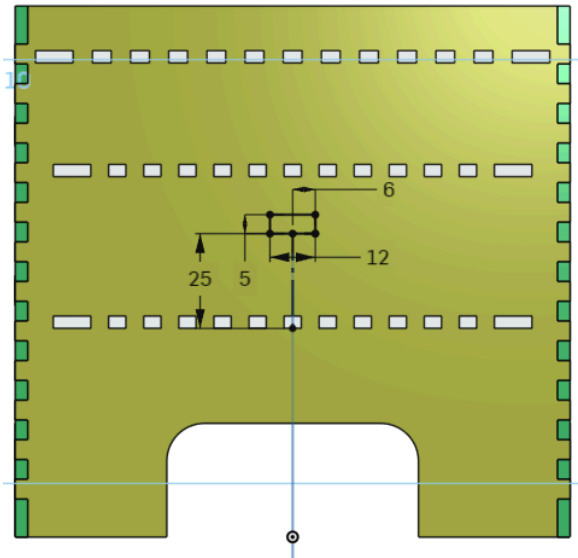


GABINETE LATERAL DERECHO: Para hacer esta pieza se usó la herramienta de simetría para duplicar el lateral izquierdo y no tener que hacer otro boceto.

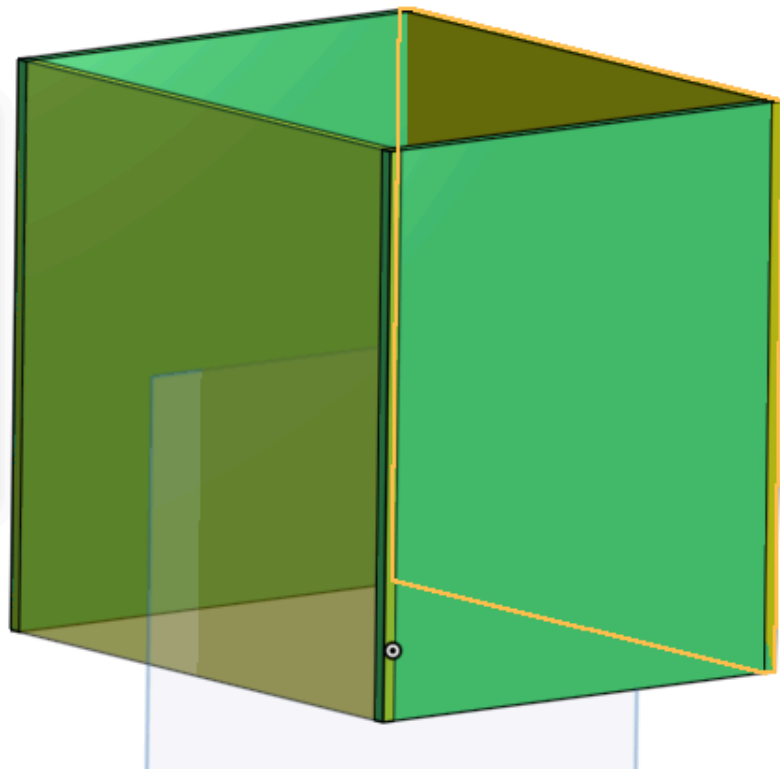
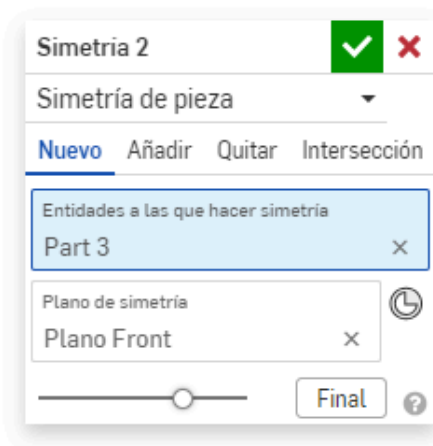


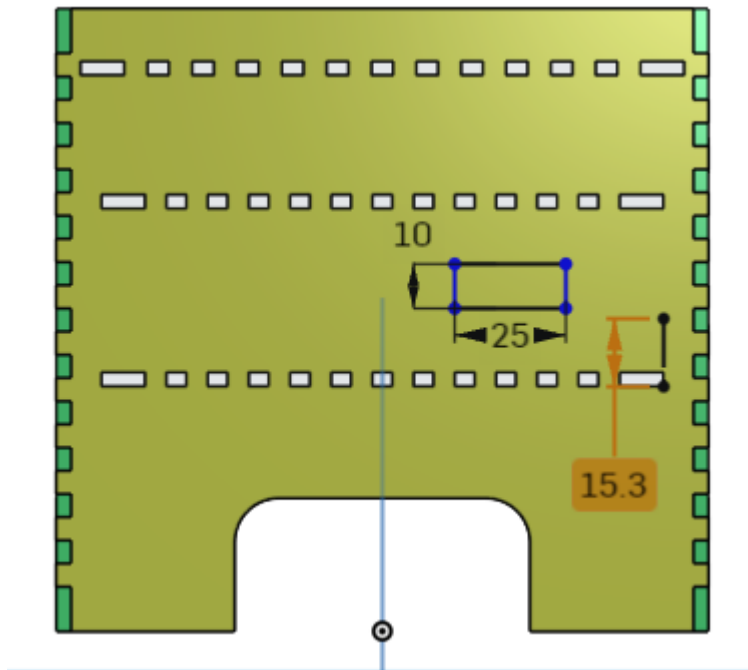
GABINETE PARTE FRONTAL: En esta pieza se hizo un agujero rectangular para poder alimentar a la placa principal(que también alimenta a la placa de control de nivel de agua) mediante un cable micro usb B, usando un cargador de celular de 5V 2A.



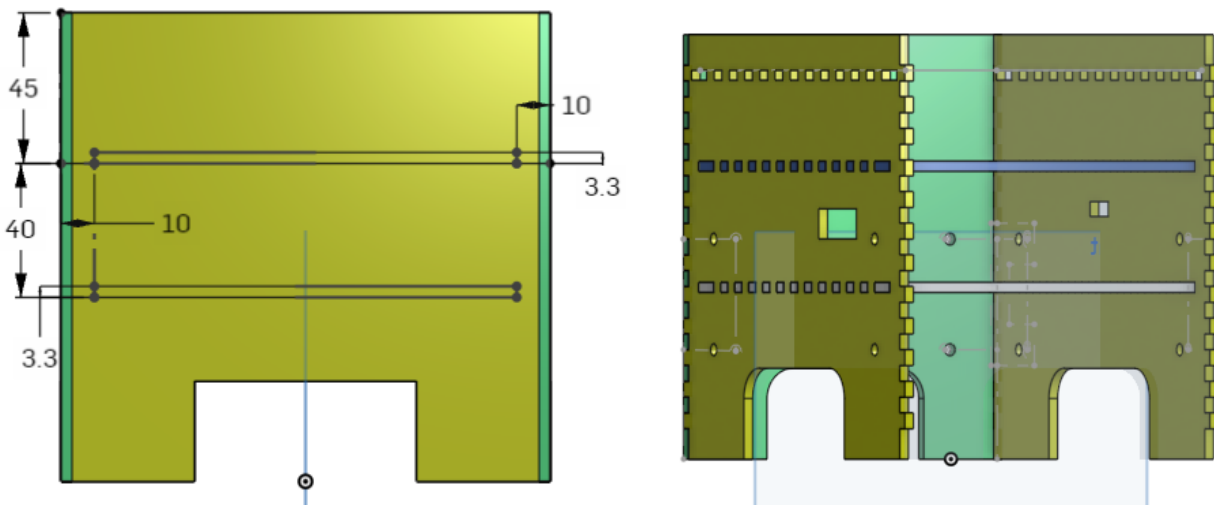


GABINETE PARTE TRASERA: Para hacer esta pieza se hizo uso también de la herramienta simetría para duplicar la parte trasera sin tener que usar otro boceto. A esta pieza se le hizo un agujero rectangular para poder hacer las conexiones de los sensores del invernadero

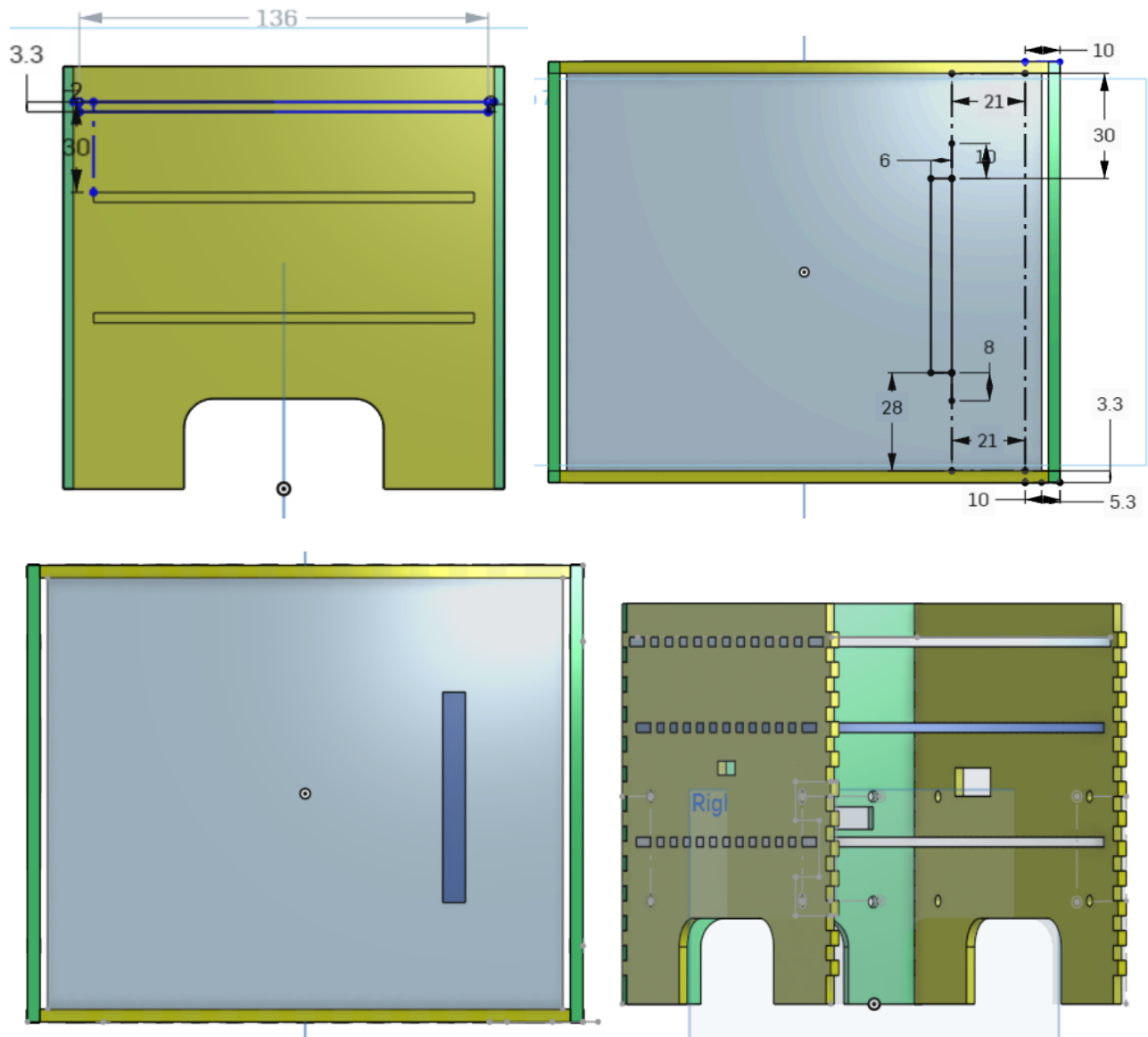




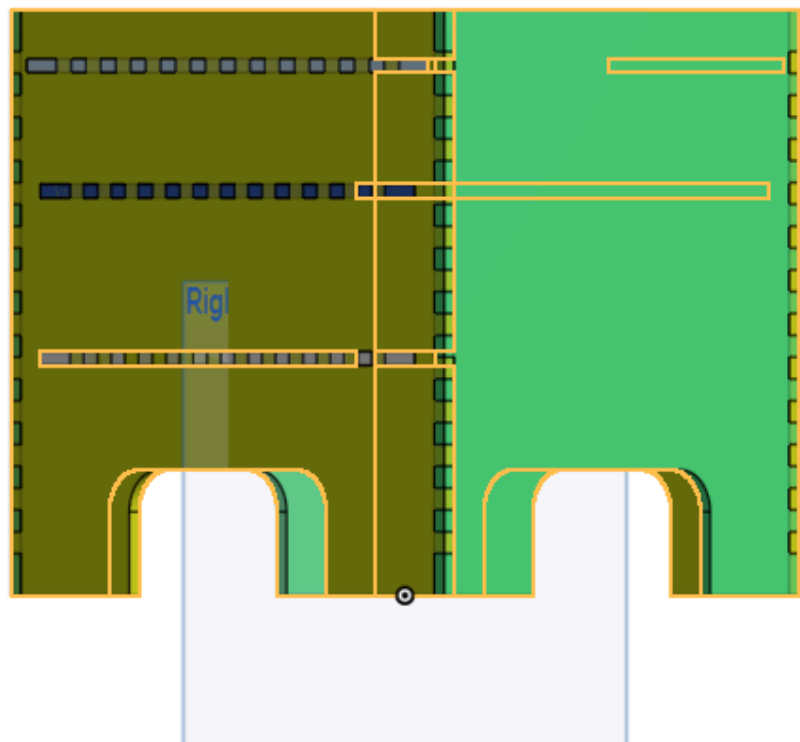
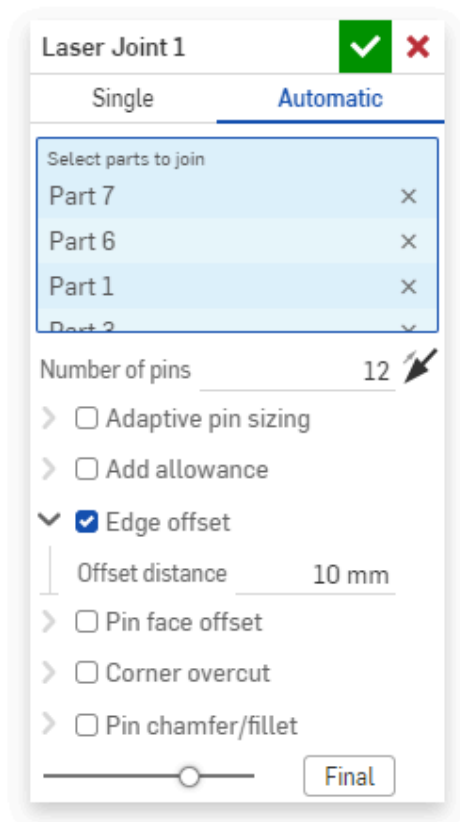
GABINETE, PLATAFORMAS PARA LAS PLACAS: Entre la parte trasera y delantera se pusieron 2 plataformas para poder anclar las placas. La superior es para la placa del control de nivel del agua y la de abajo es para la placa principal.



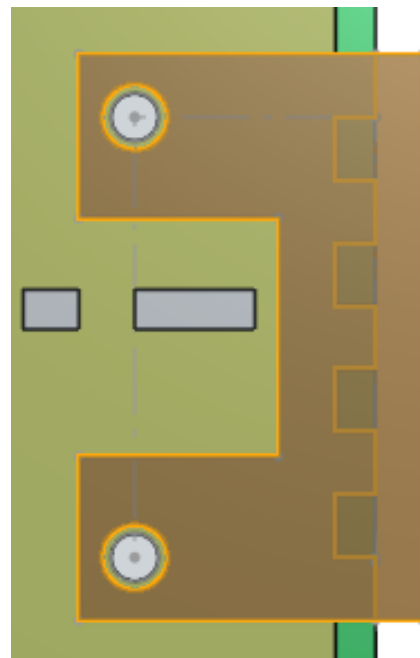
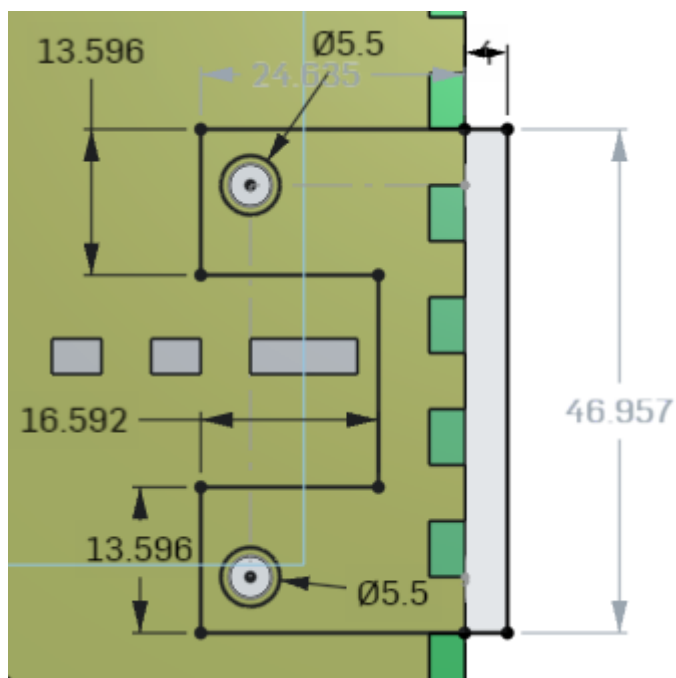
GABINETE, TAPA: Se hizo esta pieza para poder dar protección a la placa en la parte superior, además se le hizo un agujero rectangular para que se puedan ver los leds indicadores del nivel del agua.

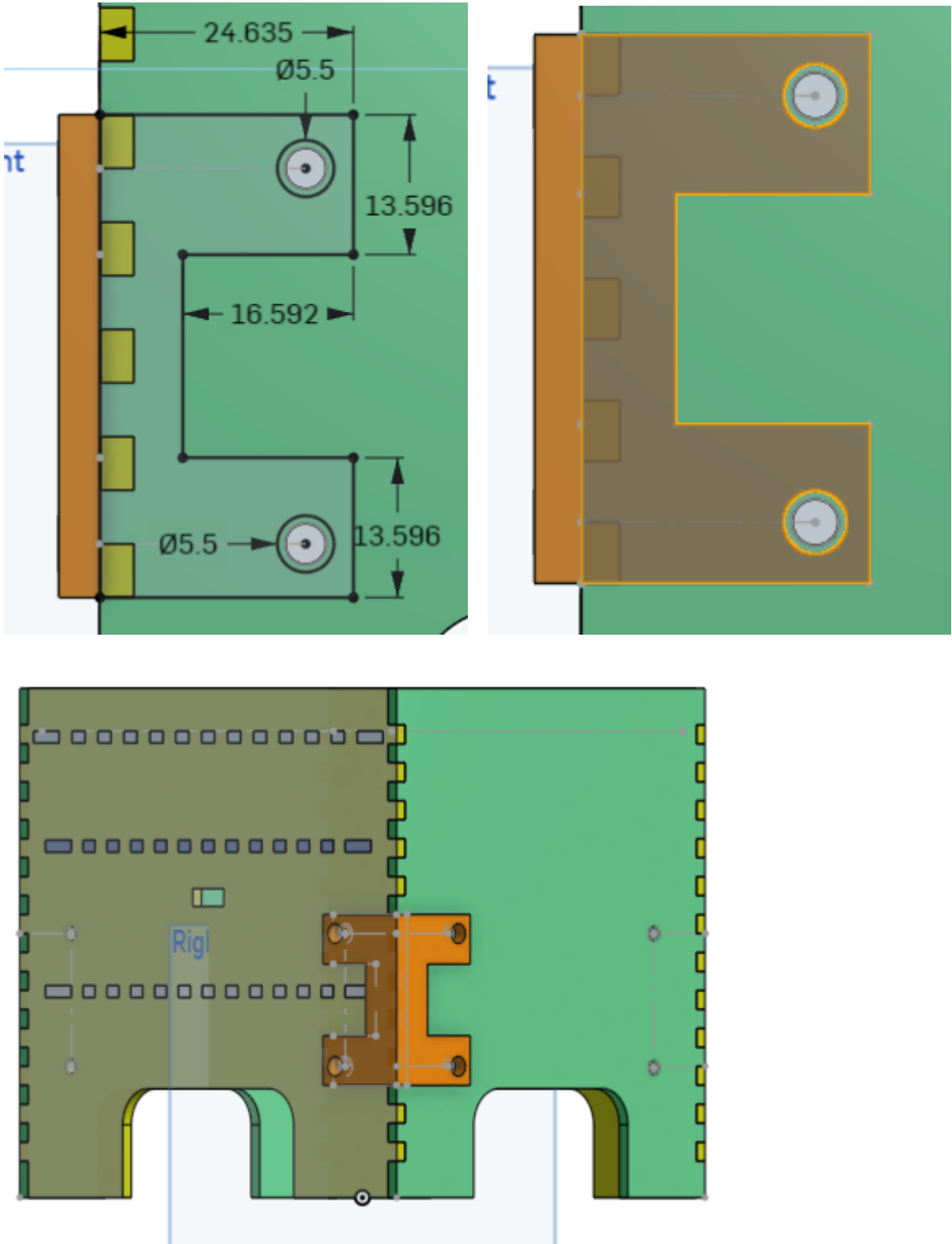


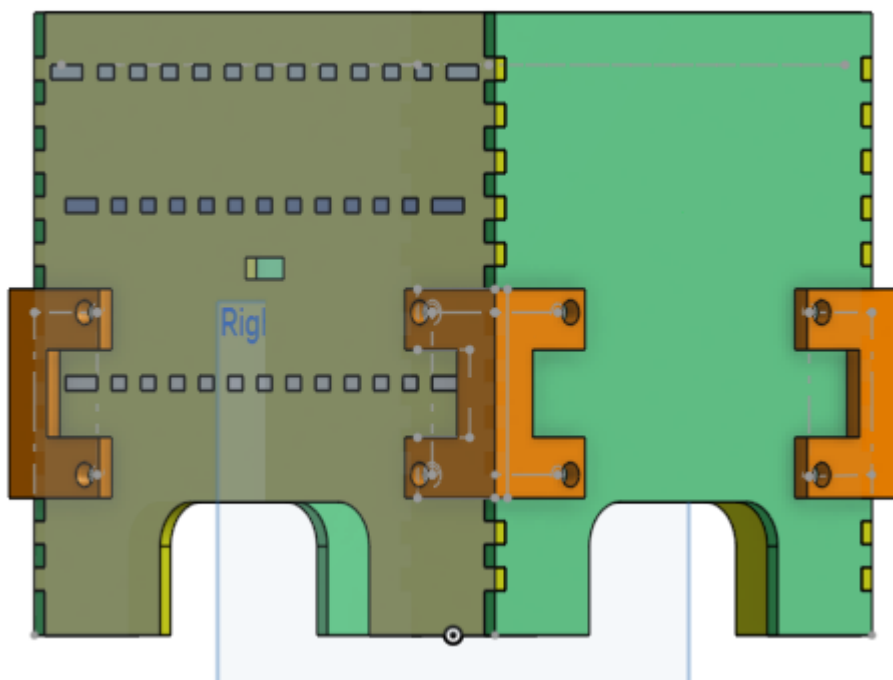
Para hacer los dientes para que las piezas se pudieran unir se hizo uso de la herramienta Laser Joint:



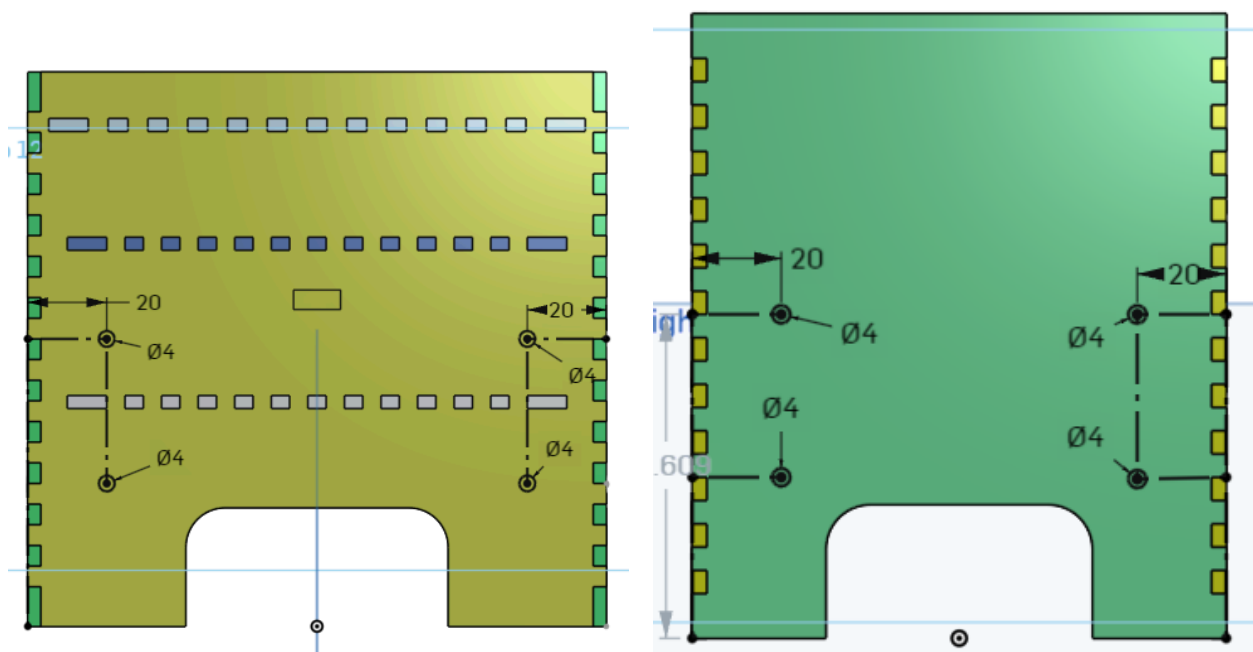
Para la unión entre las 4 piezas (delantera, trasera, lateral izquierda y lateral derecho) se hizo el diseño de unos soportes y luego se lo imprimió 4 veces con la impresora 3D, para las esquinas.







Para que los soportes se pudieran unir a las piezas con tornillos, se hizo a cada una agujeros de 4mm de diámetro



ANÁLISIS DE COSTOS

Código	Descripción	Cantidad	P.Unit	TOTAL \$ARS
ULN2803-SMD	ULN2803 SMD SOL-18	1	\$1528,35	\$1528,35
CD4049	HEX INVERTING BUFFER	1	\$1852,54	\$1852,54
ZT16A	ZÓCALO DIL 16	1	\$162,04	\$162,04
CR1206-22K	RESISTOR SMD 22K	6	\$9,26	\$55,57
CR1206-1K	RESISTOR SMD 1K	16	\$9,26	\$148,20
CR1206-10K	RESISTOR SMD 10K	6	\$9,26	\$55,57
2N3904	NPN GENERAL PURPOSE AMPLIFIER	4	\$46,31	\$92,62
XDT-126VP-2P	BORNERA 2 POLOS PASO 5MM 8AMP	8	\$157,47	\$1259,76
XDT-126VP-3P	BORNERA 3 POLOS PASO 5MM 8AMP	3	\$231,57	\$694,71
FRS10CS10-05 V	RELE 1 INVERSOR 5VDC 10ª PARA IMPRESO	2	\$1836,44	\$3672,88
L-5G	LED 5mm VERDE DIFUSO	4	\$122,29	\$489,16
L-5R	LED 5mm ROJO DIFUSO	4	\$73,99	\$295,96
L-5Y	LED 5mm AMARILLO DIFUSO	2	\$128,46	\$256,91
LDR-10	LDR 10mm 250mW 300V 600nm	4	\$2883,80	\$11535,16
TP1X40	TIRA DE PINES 1x40 MACHO P/IMPRESO	1	\$542,60	\$542,60
TPB1X40	TIRA DE PINES 1x40 90° MACHO P/IMPRESO	1	\$984,90	\$984,90
PB40S	HEMBRA .1" 40 VIAS RECTO	1	\$546.322	\$546.322
CC-211010	PLACA 10x10 VIRGEN S/FAZ FENOLICO	1	\$2081,81	\$2081,81
CC-210505	PLACA 5x5 VIRGEN S/FAZ FENOLICO	1	\$672,25	\$672,25
FDLL4148	1N4148 SMD MINI MELF (DL-35)	2	\$62,35	\$124,71
AR-DHT11 BOARD	SENSOR HUMEDAD Y TEMP C/BOARD	1	\$3107,81	\$3107,81
CAPACITIVE-MOISTURE-SENSOR-1	SENSOR DE HUMEDAD DE SUELO CON CONECTOR GRAVITY 3 PINES	1	\$3066,06	\$3066,06

TOTAL \$ARS= \$33.225,19. TOTAL U\$S= \$24,61

HOJA DE DATOS DE LOS COMPONENTES

1N4007

1N4001 THRU 1N4007

PLASTIC SILICON RECTIFIER

VOLTAGE - 50 to 1000 Volts CURRENT - 1.0 Ampere

FEATURES

- Low forward voltage drop
- High current capability
- High reliability
- High surge current capability
- Exceeds environmental standards of MIL-S-19500/228

MECHANICAL DATA

Case: Molded plastic, DO-41

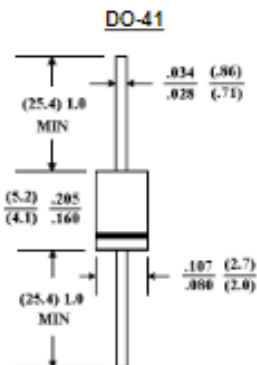
Epoxy: UL 94V-O rate flame retardant

Lead: Axial leads, solderable per MIL-STD-202, method 208 guaranteed

Polarity: Color band denotes cathode end

Mounting Position: Any

Weight: 0.012 ounce, 0.3 gram



Dimensions in inches and (millimeters)

MAXIMUM RATINGS AND ELECTRICAL CHARACTERISTICS

Ratings at 25 °C ambient temperature unless otherwise specified.

Single phase, half wave, 60 Hz, resistive or inductive load.

For capacitive load, derate current by 20%.

	1N4001	1N4002	1N4003	1N4004	1N4005	1N4006	1N4007	UNITS
Maximum Recurrent Peak Reverse Voltage	50	100	200	400	600	800	1000	V
Maximum RMS Voltage	35	75	140	280	420	560	700	V
Maximum DC Blocking Voltage	50	100	200	400	600	800	1000	V
Maximum Average Forward Rectified Current .375" (9.5mm) Lead Length at T _A =75 °C	1.0							A
Peak Forward Surge Current 8.3ms single half sine-wave superimposed on rated load (JEDEC method)	30							A
Maximum Forward Voltage at 1.0A DC and 25 °C	1.1							V
Maximum Full Load Reverse Current Full Cycle Average at 75 °C Ambient	30							µg A
Maximum Reverse Current at T _A =25 °C	5.0							µg A
At Rated DC Blocking Voltage T _A =100 °C	500							µg A
Typical Junction capacitance (Note 1)	15							pF
Typical Thermal Resistance (Note 2) R _{θKJA}	50							°C/W
Typical Thermal resistance (NOTE 2) R _{θKJL}	25							°C/W
Operating and Storage Temperature Range T _A , T _{STG}	-55 to +150							°C

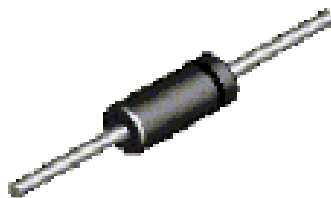
NOTES:

1. Measured at 1 MHz and applied reverse voltage of 4.0 VDC.
2. Thermal Resistance Junction to Ambient and from junction to lead at 0.375" (9.5mm) lead length P.C.B mounted.

1N4148



1N/FDLL 914/A/B / 916/A/B / 4148 / 4448



DO-35



LL-34

THE PLACEMENT OF THE EXPANSION GAP HAS NO RELATIONSHIP TO THE LOCATION OF THE CATHODE TERMINAL.

COLOR BAND MARKING		
DEVICE	1ST BAND	2ND BAND
FDLL914	BLACK	BROWN
FDLL914A	BLACK	GRAY
FDLL914B	BROWN	BLACK
FDLL916	BLACK	RED
FDLL916A	BLACK	WHITE
FDLL916B	BROWN	BROWN
FDLL4148	BLACK	BROWN
FDLL4448	BROWN	BLACK

Small Signal Diode

Absolute Maximum Ratings*

T_J = 25°C unless otherwise noted

Symbol	Parameter	Value	Units
V _{RRM}	Maximum Repetitive Reverse Voltage	100	V
I _{F(AV)}	Average Rectified Forward Current	200	mA
I _{FSM}	Non-repetitive Peak Forward Surge Current	1.0	A
	Pulse Width = 1.0 second	4.0	A
T _{stg}	Storage Temperature Range	-65 to +200	°C
T _J	Operating Junction Temperature	175	°C

* These ratings are limiting values above which the serviceability of any semiconductor device may be impaired.

NOTES:

- 1) These ratings are based on a maximum junction temperature of 200 degrees C.
2) These are steady state limits. The factory should be consulted on applications involving pulsed or low duty cycle operations.

Thermal Characteristics

Symbol	Characteristic	Max	Units
		1N/FDLL 914/A/B / 4148 / 4448	
P _D	Power Dissipation	500	mW
R _{θJA}	Thermal Resistance, Junction to Ambient	300	°C/W

Small Signal Diode
(continued)

Electrical Characteristics $T_A = 25^\circ\text{C}$ unless otherwise noted

Symbol	Parameter	Test Conditions	Min	Max	Units
V_R	Breakdown Voltage	$I_R = 100\ \mu\text{A}$ $I_R = 5.0\ \mu\text{A}$	100 75		V V
V_F	Forward Voltage	1N914B/4448 $I_F = 5.0\ \text{mA}$	620	720	mV
		1N916B $I_F = 5.0\ \text{mA}$	630	730	mV
		1N914/916/4148 $I_F = 10\ \text{mA}$		1.0	V
		1N914A/916A $I_F = 20\ \text{mA}$		1.0	V
		1N916B $I_F = 20\ \text{mA}$		1.0	V
		1N914B/4448 $I_F = 100\ \text{mA}$		1.0	V
I_R	Reverse Current	$V_R = 20\ \text{V}$		25	nA
		$V_R = 20\ \text{V}, T_A = 150^\circ\text{C}$		50	μA
		$V_R = 75\ \text{V}$		5.0	μA
C_T	Total Capacitance	1N916A/B/4448 $V_R = 0, f = 1.0\ \text{MHz}$		2.0	pF
		1N914A/B/4148 $V_R = 0, f = 1.0\ \text{MHz}$		4.0	pF
t_r	Reverse Recovery Time	$I_F = 10\ \text{mA}, V_R = 6.0\ \text{V}$ (60mA), $I_R = 1.0\ \text{mA}, R_s = 100\Omega$		4.0	ns

Typical Characteristics

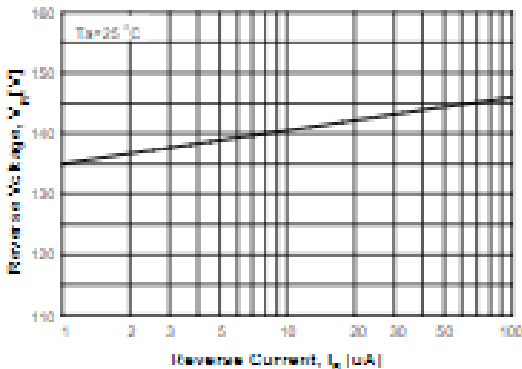


Figure 1. Reverse Voltage vs Reverse Current
BV - 1.0 to 100 μA

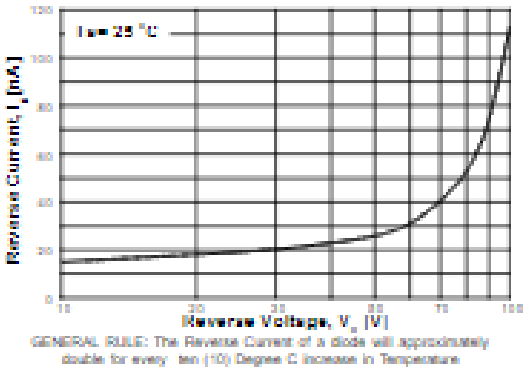


Figure 2. Reverse Current vs Reverse Voltage
IR - 10 to 100 V

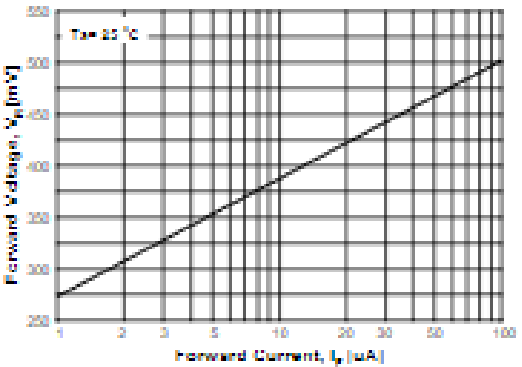


Figure 3. Forward Voltage vs Forward Current
VF - 1 to 100 μA

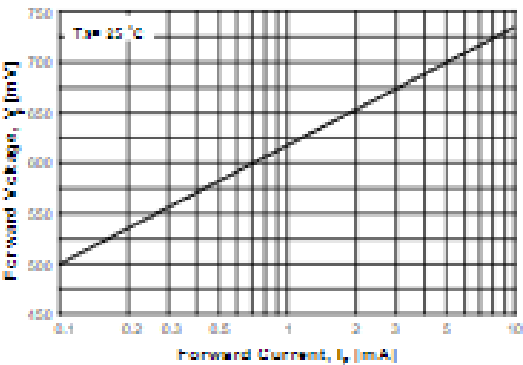


Figure 4. Forward Voltage vs Forward Current
VF - 0.1 to 10 mA

Relay 5DC

SONGLE RELAY

	RELAY ISO9002	SRD
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1. MAIN FEATURES

- Switching capacity available by 10A in spite of small size design for highdensity P.C. board mounting technique.
- UL,CUL,TUV recognized.
- Selection of plastic material for high temperature and better chemical solution performance.
- Sealed types available.
- Simple relay magnetic circuit to meet low cost of mass production.

2. APPLICATIONS

- Domestic appliance, office machine, audio, equipment, automobile, etc.
(Remote control TV receiver, monitor display, audio equipment high rushing current use application.)

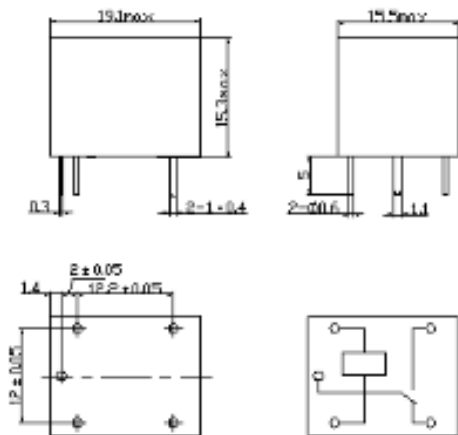
3. ORDERING INFORMATION

SRD	XX VDC	S	L	C
Model of relay	Nominal coil voltage	Structure	Coil sensitivity	Contact form
SRD	03、05、06、09、12、24、48VDC	S:Sealed type	L:0.36W	A:1 form A
		F:Flux free type	D:0.45W	B:1 form B
				C:1 form C

4. RATING

CCC	FILE NUMBER:CH0052885-2000	7A/240VDC
CCC	FILE NUMBER:CH0036746-99	10A/250VDC
UL/CUL	FILE NUMBER: E167996	10A/125VAC 28VDC
TUV	FILE NUMBER: R9933789	10A/240VAC 28VDC

5. DIMENSION(unit:mm) DRILLING(unit:mm) WIRING DIAGRAM



6. COIL DATA CHART (AT20°C)

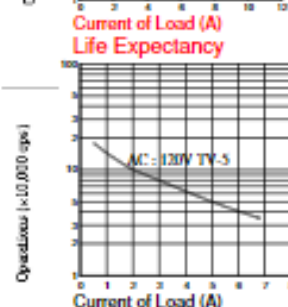
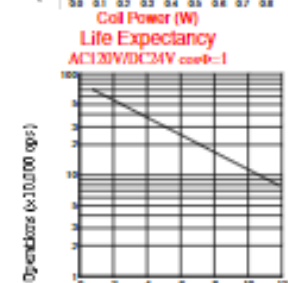
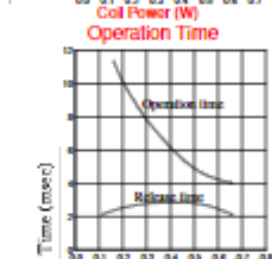
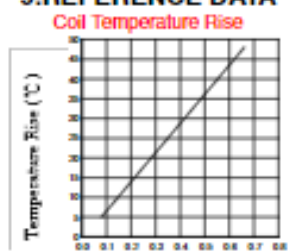
Coil Sensitivity	Coil Voltage Code	Nominal Voltage (VDC)	Nominal Current (mA)	Coil Resistance (Ω) $\pm 10\%$	Power Consumption (W)	Pull-In Voltage (VDC)	Drop-Out Voltage (VDC)	Max-Allowable Voltage (VDC)
SRD (High Sensitivity)	03	03	120	25	abt. 0.36W	75%Max.	10% Min.	120%
	05	05	71.4	70				
	06	06	60	100				
	09	09	40	225				
	12	12	30	400				
	24	24	15	1600				
SRD (Standard)	03	03	150	20	abt. 0.45W	75% Max.	10% Min.	110%
	05	05	89.3	55				
	06	06	75	80				
	09	09	50	180				
	12	12	37.5	320				
	24	24	18.7	1280				
	48	48	10	4500	abt. 0.51W			

7. CONTACT RATING

Item	Type	SRD
	FORM C	FORM A
Contact Capacity	7A 28VDC	10A 28VDC
Resistive Load ($\cos\phi=1$)	10A 125VAC	10A 240VAC
	7A 240VAC	
Inductive Load ($\cos\phi=0.4$ L/R=7msec)	3A 120VAC	5A 120VAC
	3A 28VDC	5A 28VDC
Max. Allowable Voltage	250VAC/110VDC	250VAC/110VDC
Max. Allowable Power Force	800VAC/240W	1200VA/300W
Contact Material	AgCdO	AgCdO

8. PERFORMANCE (at initial value)

Item	Type	SRD
Contact Resistance		100m Ω Max.
Operation Time		10msec Max.
Release Time		5msec Max.
Dielectric Strength	Between coil & contact	1500VAC 50/60HZ (1 minute)
	Between contacts	1000VAC 50/60HZ (1 minute)
Insulation Resistance		100 M Ω Min. (500VDC)
Max. ON/OFF Switching		
Mechanically		300 operation/min
Electrically		30 operation/min
Ambient Temperature		-25°C to +70°C
Operating Humidity		45 to 85% RH
Vibration		
Endurance		10 to 55Hz Double Amplitude 1.5mm
Error Operation		10 to 55Hz Double Amplitude 1.5mm
Shock		
Endurance		100G Min.
Error Operation		10G Min.
Life Expectancy		
Mechanically		10 ⁷ operations. Min. (no load)
Electrically		10 ⁵ operations. Min. (at rated coil voltage)
Weight		abt. 10grs.

9. REFERENCE DATA

2N3904

2N3903, 2N3904

2N3903 is a Preferred Device

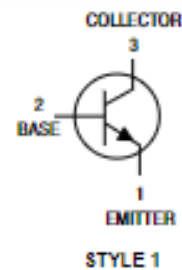
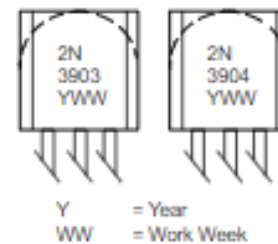
General Purpose Transistors**NPN Silicon****MAXIMUM RATINGS**

Rating	Symbol	Value	Unit
Collector-Emitter Voltage	V_{CE0}	40	Vdc
Collector-Base Voltage	V_{CB0}	60	Vdc
Emitter-Base Voltage	V_{EB0}	6.0	Vdc
Collector Current – Continuous	I_C	200	mA dc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	625 5.0	mW mW/ $^\circ\text{C}$
Total Device Dissipation @ $T_C = 25^\circ\text{C}$ Derate above 25°C	P_D	1.5 12	Watts mW/ $^\circ\text{C}$
Operating and Storage Junction Temperature Range	T_J, T_{stg}	-55 to +150	$^\circ\text{C}$

THERMAL CHARACTERISTICS (Note 1.)

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	200	$^\circ\text{C/W}$
Thermal Resistance, Junction to Case	$R_{\theta JC}$	83.3	$^\circ\text{C/W}$

1. Indicates Data in addition to JEDEC Requirements.

**ON Semiconductor™**<http://onsemi.com>**MARKING DIAGRAMS****ORDERING INFORMATION**

Device	Package	Shipping
2N3903	TO-92	5000 Units/Box
2N3903RLRM	TO-92	2000/Ammo Pack
2N3904	TO-92	5000 Units/Box
2N3904RLRA	TO-92	2000/Tape & Reel
2N3904RLRE	TO-92	2000/Tape & Reel
2N3904RLRM	TO-92	2000/Ammo Pack
2N3904RLRP	TO-92	2000/Ammo Pack
2N3904RL1	TO-92	2000/Tape & Reel
2N3904ZL1	TO-92	2000/Ammo Pack

Preferred devices are recommended choices for future use and best overall value.

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Characteristic	Symbol	Min	Max	Unit
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OFF CHARACTERISTICS

Collector-Emitter Breakdown Voltage (Note 2.) ($I_C = 1.0\text{ mA}$, $I_B = 0$)	$V_{(BR)CEO}$	40	—	Vdc
Collector-Base Breakdown Voltage ($I_C = 10\text{ }\mu\text{A}$, $I_E = 0$)	$V_{(BR)CBO}$	60	—	Vdc
Emitter-Base Breakdown Voltage ($I_E = 10\text{ }\mu\text{A}$, $I_C = 0$)	$V_{(BR)EBO}$	6.0	—	Vdc
Base Cutoff Current ($V_{CE} = 30\text{ Vdc}$, $V_{EB} = 3.0\text{ Vdc}$)	I_{BL}	—	50	nAdc
Collector Cutoff Current ($V_{CE} = 30\text{ Vdc}$, $V_{EB} = 3.0\text{ Vdc}$)	I_{CEX}	—	50	nAdc

ON CHARACTERISTICS

DC Current Gain (Note 2.) ($I_C = 0.1\text{ mA}$, $V_{CE} = 1.0\text{ Vdc}$)	2N3903 2N3904	h_{FE}	20 40	—	—
($I_C = 1.0\text{ mA}$, $V_{CE} = 1.0\text{ Vdc}$)	2N3903 2N3904		35 70	—	—
($I_C = 10\text{ mA}$, $V_{CE} = 1.0\text{ Vdc}$)	2N3903 2N3904		50 100	150 300	—
($I_C = 50\text{ mA}$, $V_{CE} = 1.0\text{ Vdc}$)	2N3903 2N3904		30 60	—	—
($I_C = 100\text{ mA}$, $V_{CE} = 1.0\text{ Vdc}$)	2N3903 2N3904		15 30	—	—
Collector-Emitter Saturation Voltage (Note 2.) ($I_C = 10\text{ mA}$, $I_B = 1.0\text{ mA}$) ($I_C = 50\text{ mA}$, $I_B = 5.0\text{ mA}$)		$V_{CE(sat)}$	— —	0.2 0.3	Vdc
Base-Emitter Saturation Voltage (Note 2.) ($I_C = 10\text{ mA}$, $I_B = 1.0\text{ mA}$) ($I_C = 50\text{ mA}$, $I_B = 5.0\text{ mA}$)		$V_{BE(sat)}$	0.65 —	0.85 0.95	Vdc

SMALL-SIGNAL CHARACTERISTICS

Current-Gain – Bandwidth Product ($I_C = 10\text{ mA}$, $V_{CE} = 20\text{ Vdc}$, $f = 100\text{ MHz}$)	2N3903 2N3904	f_T	250 300	—	MHz
Output Capacitance ($V_{CB} = 5.0\text{ Vdc}$, $I_E = 0$, $f = 1.0\text{ MHz}$)		C_{obo}	—	4.0	pF
Input Capacitance ($V_{EB} = 0.5\text{ Vdc}$, $I_C = 0$, $f = 1.0\text{ MHz}$)		C_{ibo}	—	8.0	pF
Input Impedance ($I_C = 1.0\text{ mA}$, $V_{CE} = 10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	2N3903 2N3904	h_{ie}	1.0 1.0	8.0 10	k Ω
Voltage Feedback Ratio ($I_C = 1.0\text{ mA}$, $V_{CE} = 10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	2N3903 2N3904	h_{re}	0.1 0.5	5.0 8.0	$\times 10^{-4}$
Small-Signal Current Gain ($I_C = 1.0\text{ mA}$, $V_{CE} = 10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	2N3903 2N3904	h_{fe}	50 100	200 400	—
Output Admittance ($I_C = 1.0\text{ mA}$, $V_{CE} = 10\text{ Vdc}$, $f = 1.0\text{ kHz}$)		h_{oe}	1.0	40	μmhos
Noise Figure ($I_C = 100\text{ }\mu\text{A}$, $V_{CE} = 5.0\text{ Vdc}$, $R_S = 1.0\text{ k}\Omega$, $f = 1.0\text{ kHz}$)	2N3903 2N3904	NF	— —	6.0 5.0	dB

SWITCHING CHARACTERISTICS

Delay Time	$(V_{CC} = 3.0\text{ Vdc}$, $V_{BE} = 0.5\text{ Vdc}$, $I_C = 10\text{ mA}$, $I_{B1} = 1.0\text{ mA}$)	t_d	—	35	ns
Rise Time		t_r	—	35	ns
Storage Time	$(V_{CC} = 3.0\text{ Vdc}$, $I_C = 10\text{ mA}$, $I_{B1} = I_{B2} = 1.0\text{ mA}$)	t_s	—	175 200	ns
Fall Time		t_f	—	50	ns

2. Pulse Test: Pulse Width $\leq 300\text{ }\mu\text{s}$; Duty Cycle $\leq 2\%$.

ULN2803



MOTOROLA

Octal High Voltage, High Current Darlington Transistor Arrays

The eight NPN Darlington connected transistors in this family of arrays are ideally suited for interfacing between low logic level digital circuitry (such as TTL, CMOS or PMOS/NMOS) and the higher current/voltage requirements of lamps, relays, printer hammers or other similar loads for a broad range of computer, industrial, and consumer applications. All devices feature open-collector outputs and free wheeling clamp diodes for transient suppression.

The ULN2803 is designed to be compatible with standard TTL families while the ULN2804 is optimized for 6 to 15 volt high level CMOS or PMOS.

MAXIMUM RATINGS ($T_A = 25^\circ\text{C}$ and rating apply to any one device in the package, unless otherwise noted.)

Rating	Symbol	Value	Unit
Output Voltage	V_O	50	V
Input Voltage (Except ULN2801)	V_I	30	V
Collector Current – Continuous	I_C	500	mA
Base Current – Continuous	I_B	25	mA
Operating Ambient Temperature Range	T_A	0 to +70	$^\circ\text{C}$
Storage Temperature Range	T_{stg}	-55 to +150	$^\circ\text{C}$
Junction Temperature	T_J	125	$^\circ\text{C}$

$R_{\theta JA} = 55^\circ\text{C/W}$

Do not exceed maximum current limit per driver.

ORDERING INFORMATION

Device	Characteristics		
	Input Compatibility	$V_{CE}(\text{Max})/I_C(\text{Max})$	Operating Temperature Range
ULN2803A	TTL, 5.0 V CMOS	50 V/500 mA	$T_A = 0 \text{ to } +70^\circ\text{C}$
ULN2804A	6 to 15 V CMOS, PMOS		

Order this document by ULN2803/D

ULN2803 ULN2804

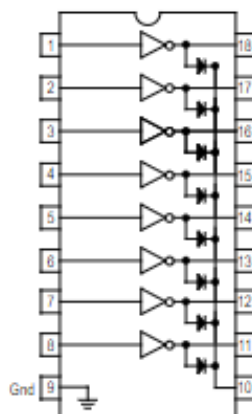
OCTAL PERIPHERAL DRIVER ARRAYS

SEMICONDUCTOR TECHNICAL DATA



A SUFFIX
PLASTIC PACKAGE
CASE 707

PIN CONNECTIONS



▼ Motorola, Inc. 1996

Rev 1

ULN2803 ULN2804**ELECTRICAL CHARACTERISTICS** ($T_A = 25^\circ\text{C}$, unless otherwise noted)

Characteristic		Symbol	Min	Typ	Max	Unit
Output Leakage Current (Figure 1) ($V_O = 50\text{ V}$, $T_A = +70^\circ\text{C}$) ($V_O = 50\text{ V}$, $T_A = +25^\circ\text{C}$) ($V_O = 50\text{ V}$, $T_A = +70^\circ\text{C}$, $V_I = 6.0\text{ V}$) ($V_O = 50\text{ V}$, $T_A = +70^\circ\text{C}$, $V_I = 1.0\text{ V}$)	All Types All Types ULN2802 ULN2804	I_{CEX}	– – – –	– – – –	100 50 500 500	μA
Collector-Emitter Saturation Voltage (Figure 2) ($I_C = 350\text{ mA}$, $I_B = 500\text{ }\mu\text{A}$) ($I_C = 200\text{ mA}$, $I_B = 350\text{ }\mu\text{A}$) ($I_C = 100\text{ mA}$, $I_B = 250\text{ }\mu\text{A}$)	All Types All Types All Types	$V_{CE(sat)}$	– – –	1.1 0.95 0.85	1.6 1.3 1.1	V
Input Current – On Condition (Figure 4) ($V_I = 17\text{ V}$) ($V_I = 3.85\text{ V}$) ($V_I = 5.0\text{ V}$) ($V_I = 12\text{ V}$)	ULN2802 ULN2803 ULN2804 ULN2804	$I_{I(on)}$	– – – –	0.82 0.93 0.35 1.0	1.25 1.35 0.5 1.45	mA
Input Voltage – On Condition (Figure 5) ($V_{CE} = 2.0\text{ V}$, $I_C = 300\text{ mA}$) ($V_{CE} = 2.0\text{ V}$, $I_C = 200\text{ mA}$) ($V_{CE} = 2.0\text{ V}$, $I_C = 250\text{ mA}$) ($V_{CE} = 2.0\text{ V}$, $I_C = 300\text{ mA}$) ($V_{CE} = 2.0\text{ V}$, $I_C = 125\text{ mA}$) ($V_{CE} = 2.0\text{ V}$, $I_C = 200\text{ mA}$) ($V_{CE} = 2.0\text{ V}$, $I_C = 275\text{ mA}$) ($V_{CE} = 2.0\text{ V}$, $I_C = 350\text{ mA}$)	ULN2802 ULN2803 ULN2803 ULN2803 ULN2804 ULN2804 ULN2804 ULN2804	$V_{I(on)}$	– – – – – – – –	– – – – – – – –	13 2.4 2.7 3.0 5.0 6.0 7.0 8.0	V
Input Current – Off Condition (Figure 3) ($I_C = 500\text{ }\mu\text{A}$, $T_A = +70^\circ\text{C}$)	All Types	$I_{I(off)}$	50	100	–	μA
DC Current Gain (Figure 2) ($V_{CE} = 2.0\text{ V}$, $I_C = 350\text{ mA}$)	ULN2801	h_{FE}	1000	–	–	–
Input Capacitance		C_I	–	15	25	pF
Turn-On Delay Time (50% E_I to 50% E_O)		t_{on}	–	0.25	1.0	μs
Turn-Off Delay Time (50% E_I to 50% E_O)		t_{off}	–	0.25	1.0	μs
Clamp Diode Leakage Current (Figure 6) ($V_R = 50\text{ V}$)	$T_A = +25^\circ\text{C}$ $T_A = +70^\circ\text{C}$	I_R	–	–	50 100	μA
Clamp Diode Forward Voltage (Figure 7) ($I_F = 350\text{ mA}$)		V_F	–	1.5	2.0	V

ESP32-WROOM-32

1. Overview

ESP32-WROOM-32 is a powerful, generic Wi-Fi+BT+BLE MCU module that targets a wide variety of applications, ranging from low-power sensor networks to the most demanding tasks, such as voice encoding, music streaming and MP3 decoding.

At the core of this module is the ESP32-D0WDQ6 chip*. The chip embedded is designed to be scalable and adaptive. There are two CPU cores that can be individually controlled, and the CPU clock frequency is adjustable from 80 MHz to 240 MHz. The user may also power off the CPU and make use of the low-power co-processor to constantly monitor the peripherals for changes or crossing of thresholds. ESP32 integrates a rich set of peripherals, ranging from capacitive touch sensors, Hall sensors, SD card interface, Ethernet, high-speed SPI, UART, PS and PC.

Note:

* For details on the part numbers of the ESP32 family of chips, please refer to the document [ESP32 Datasheet](#).

The integration of Bluetooth, Bluetooth LE and Wi-Fi ensures that a wide range of applications can be targeted, and that the module is all-around: using Wi-Fi allows a large physical range and direct connection to the Internet through a Wi-Fi router, while using Bluetooth allows the user to conveniently connect to the phone or broadcast low energy beacons for its detection. The sleep current of the ESP32 chip is less than 5 μ A, making it suitable for battery powered and wearable electronics applications. The module supports a data rate of up to 150 Mbps, and 20 dBm output power at the antenna to ensure the widest physical range. As such the module does offer industry-leading specifications and the best performance for electronic integration, range, power consumption, and connectivity.

The operating system chosen for ESP32 is freeRTOS with LwIP; TLS 1.2 with hardware acceleration is built in as well. Secure (encrypted) over the air (OTA) upgrade is also supported, so that users can upgrade their products even after their release, at minimum cost and effort.

Table 1 provides the specifications of ESP32-WROOM-32.

Table 1: ESP32-WROOM-32 Specifications

Categories	Items	Specifications
Certification	RF certification	FCC/CE-RED/IC/TELEC/KCC/SRRC/NCC
	Wi-Fi certification	Wi-Fi Alliance
	Bluetooth certification	BQB
	Green certification	RoHS/REACH
Test	Reliability	HTOL/HTSL/uHAST/TCT/ESD
Wi-Fi	Protocols	802.11 b/g/n (802.11n up to 150 Mbps) A-MPDU and A-MSDU aggregation and 0.4 μ s guard interval support
	Frequency range	2.4 GHz ~ 2.5 GHz
Bluetooth	Protocols	Bluetooth v4.2 BR/EDR and BLE specification
	Radio	NZIF receiver with -97 dBm sensitivity
		Class-1, class-2 and class-3 transmitter
		AFH

Categories	Items	Specifications
	Audio	CVSD and SBC
Hardware	Module interfaces	SD card, UART, SPI, SDIO, I ² C, LED PWM, Motor PWM, I ² S, IR, pulse counter, GPIO, capacitive touch sensor, ADC, DAC
	On-chip sensor	Hall sensor
	Integrated crystal	40 MHz crystal
	Integrated SPI flash	4 MB
	Operating voltage/Power supply	2.7 V ~ 3.6 V
	Operating current	Average: 80 mA
	Minimum current delivered by power supply	500 mA
	Recommended operating temperature range	-40 °C ~ +85 °C
	Package size	(18.00±0.10) mm × (25.50±0.10) mm × (3.10±0.10) mm

2. Pin Definitions

2.1 Pin Layout

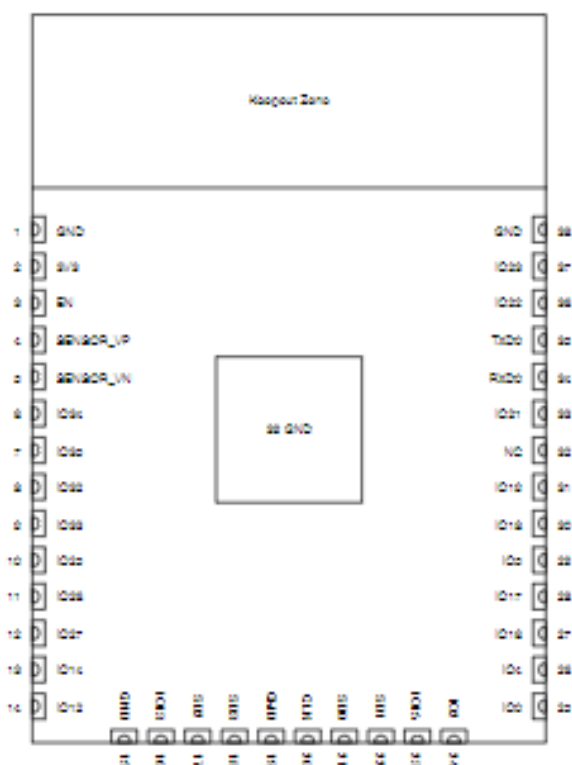


Figure 1: ESP32-WROOM-32 Pin Layout (Top View)

2.2 Pin Description

ESP32-WROOM-32 has 38 pins. See pin definitions in Table 2.

Table 2: Pin Definitions

Name	No.	Type	Function
GND	1	P	Ground
3V3	2	P	Power supply
EN	3	I	Module-enable signal. Active high.
SENSOR_VP	4	I	GPIO36, ADC1_CH0, RTC_GPIO0
SENSOR_VN	5	I	GPIO39, ADC1_CH3, RTC_GPIO3
IO34	6	I	GPIO34, ADC1_CH6, RTC_GPIO4
IO35	7	I	GPIO35, ADC1_CH7, RTC_GPIO5
IO32	8	I/O	GPIO32, XTAL_32K_P (32.768 kHz crystal oscillator input), ADC1_CH4, TOUCH9, RTC_GPIO9
IO33	9	I/O	GPIO33, XTAL_32K_N (32.768 kHz crystal oscillator output), ADC1_CH5, TOUCH8, RTC_GPIO8

Name	No.	Type	Function
IO25	10	I/O	GPIO25, DAC_1, ADC2_CH8, RTC_GPIO6, EMAC_RXD0
IO26	11	I/O	GPIO26, DAC_2, ADC2_CH9, RTC_GPIO7, EMAC_RXD1
IO27	12	I/O	GPIO27, ADC2_CH7, TOUCH7, RTC_GPIO17, EMAC_RX_DV
IO14	13	I/O	GPIO14, ADC2_CH6, TOUCH6, RTC_GPIO16, MTMS, HSPICLK, HS2_CLK, SD_CLK, EMAC_TXD2
IO12	14	I/O	GPIO12, ADC2_CH5, TOUCH5, RTC_GPIO15, MTDI, HSPIQ, HS2_DATA2, SD_DATA2, EMAC_TXD3
GND	15	P	Ground
IO13	16	I/O	GPIO13, ADC2_CH4, TOUCH4, RTC_GPIO14, MTCK, HSPID, HS2_DATA3, SD_DATA3, EMAC_RX_ER
SHD/SD2*	17	I/O	GPIO9, SD_DATA2, SPIHD, HS1_DATA2, U1RXD
SWP/SD3*	18	I/O	GPIO10, SD_DATA3, SPWP, HS1_DATA3, U1TXD
SCS/CMD*	19	I/O	GPIO11, SD_CMD, SPICSD, HS1_CMD, U1RTS
SCK/CLK*	20	I/O	GPIO6, SD_CLK, SPICLK, HS1_CLK, U1CTS
SDO/SD0*	21	I/O	GPIO7, SD_DATA0, SPIQ, HS1_DATA0, U2RTS
SDI/SD1*	22	I/O	GPIO8, SD_DATA1, SPID, HS1_DATA1, U2CTS
IO15	23	I/O	GPIO15, ADC2_CH3, TOUCH3, MTD0, HSPICSD, RTC_GPIO13, HS2_CMD, SD_CMD, EMAC_RXD3
IO2	24	I/O	GPIO2, ADC2_CH2, TOUCH2, RTC_GPIO12, HSPIWP, HS2_DATA0, SD_DATA0
IO0	25	I/O	GPIO0, ADC2_CH1, TOUCH1, RTC_GPIO11, CLK_OUT1, EMAC_TX_CLK
IO4	26	I/O	GPIO4, ADC2_CH0, TOUCH0, RTC_GPIO10, HSPIHD, HS2_DATA1, SD_DATA1, EMAC_TX_ER
IO16	27	I/O	GPIO16, HS1_DATA4, U2RXD, EMAC_CLK_OUT
IO17	28	I/O	GPIO17, HS1_DATA5, U2TXD, EMAC_CLK_OUT_180
IO5	29	I/O	GPIO5, VSPICSD, HS1_DATA6, EMAC_RX_CLK
IO18	30	I/O	GPIO18, VSPICLK, HS1_DATA7
IO19	31	I/O	GPIO19, VSPIQ, U0CTS, EMAC_TXD0
NC	32	-	-
IO21	33	I/O	GPIO21, VSPIHD, EMAC_TX_EN
RXD0	34	I/O	GPIO3, U0RXD, CLK_OUT2
TXD0	35	I/O	GPIO1, U0TXD, CLK_OUT3, EMAC_RXD2
IO22	36	I/O	GPIO22, VSPWP, U0RTS, EMAC_TXD1
IO23	37	I/O	GPIO23, VSPID, HS1_STROBE
GND	38	P	Ground

Notice:

* Pins SCK/CLK, SDO/SD0, SDI/SD1, SHD/SD2, SWP/SD3 and SCS/CMD, namely, GPIO6 to GPIO11 are connected to the integrated SPI flash integrated on the module and are not recommended for other uses.

5. Electrical Characteristics

5.1 Absolute Maximum Ratings

Stresses beyond the absolute maximum ratings listed in Table 4 below may cause permanent damage to the device. These are stress ratings only, and do not refer to the functional operation of the device that should follow the recommended operating conditions.

Table 4: Absolute Maximum Ratings

Symbol	Parameter	Min	Max	Unit
VDD33	Power supply voltage	-0.3	3.6	V
I_{output}^1	Cumulative IO output current	-	1,100	mA
T_{store}	Storage temperature	-40	150	°C

1. The module worked properly after a 24-hour test in ambient temperature at 25 °C, and the IOs in three domains (VDD3P3_RTC, VDD3P3_CPU, VDD_SDIO) output high logic level to ground. Please note that pins occupied by flash and/or PSRAM in the VDD_SDIO power domain were excluded from the test.
2. Please see Appendix IO_MUX of [ESP32 Datasheet](#) for IO's power domain.

5.2 Recommended Operating Conditions

Table 5: Recommended Operating Conditions

Symbol	Parameter	Min	Typical	Max	Unit
VDD33	Power supply voltage	2.7	3.3	3.6	V
I_{VDD}	Current delivered by external power supply	0.5	-	-	A
T	Operating temperature	-40	-	85	°C

5.3 DC Characteristics (3.3 V, 25 °C)

Table 6: DC Characteristics (3.3 V, 25 °C)

Symbol	Parameter		Min	Typ	Max	Unit
C_{IN}	Pin capacitance		-	2	-	pF
V_{IH}	High-level input voltage		$0.75 \times V_{\text{DD}}^1$	-	$V_{\text{DD}}^1 + 0.3$	V
V_{IL}	Low-level input voltage		-0.3	-	$0.25 \times V_{\text{DD}}^1$	V
I_{IH}	High-level input current		-	-	50	nA
I_{IL}	Low-level input current		-	-	50	nA
V_{OH}	High-level output voltage		$0.8 \times V_{\text{DD}}^1$	-	-	V
V_{OL}	Low-level output voltage		-	-	$0.1 \times V_{\text{DD}}^1$	V
I_{OH}	High-level source current ($V_{\text{DD}}^1 = 3.3 \text{ V}$, $V_{\text{OH}} \geq 2.64 \text{ V}$, output drive strength set to the maximum)	VDD3P3_CPU power domain ^{1, 2}	-	40	-	mA
		VDD3P3_RTC power domain ^{1, 2}	-	40	-	mA
		VDD_SDIO power domain ^{1, 3}	-	20	-	mA

Symbol	Parameter	Min	Typ	Max	Unit
I_{OL}	Low-level sink current ($V_{DD}^1 = 3.3\text{ V}$, $V_{OL} = 0.495\text{ V}$, output drive strength set to the maximum)	-	28	-	mA
R_{PU}	Resistance of internal pull-up resistor	-	45	-	k Ω
R_{PD}	Resistance of internal pull-down resistor	-	45	-	k Ω
V_{IL_nRST}	Low-level input voltage of CHIP_PU to power off the chip	-	-	0.6	V

Notes:

1. Please see Appendix IO_MUX of [ESP32 Datasheet](#) for IO's power domain. VDD is the IO voltage for a particular power domain of pins.
2. For VDD3P3_CPU and VDD3P3_RTC power domain, per-pin current sourced in the same domain is gradually reduced from around 40 mA to around 29 mA, $V_{DD} > 2.64\text{ V}$, as the number of current-source pins increases.
3. Pins occupied by flash and/or PSRAM in the VDD_SDIO power domain were excluded from the test.

5.4 Wi-Fi Radio

Table 7: Wi-Fi Radio Characteristics

Parameter	Condition	Min	Typical	Max	Unit
Input frequency	-	2412	-	2484	MHz
Output impedance*	-	-	*	-	Ω
TX power	11n, MCS7	12	13	14	dBm
	11b mode	17.5	18.5	20	dBm
Sensitivity	11b, 1 Mbps	-	-98	-	dBm
	11b, 11 Mbps	-	-89	-	dBm
	11g, 6 Mbps	-	-92	-	dBm
	11g, 54 Mbps	-	-74	-	dBm
	11n, HT20, MCS0	-	-91	-	dBm
	11n, HT20, MCS7	-	-71	-	dBm
	11n, HT40, MCS0	-	-89	-	dBm
	11n, HT40, MCS7	-	-69	-	dBm
Adjacent channel rejection	11g, 6 Mbps	-	31	-	dB
	11g, 54 Mbps	-	14	-	dB
	11n, HT20, MCS0	-	31	-	dB
	11n, HT20, MCS7	-	13	-	dB

*For the modules that use IPEX antennas, the output impedance is 50 Ω . For other modules without IPEX antennas, users do not need to concern about the output impedance.

5.5 BLE Radio

5.5.1 Receiver

Table 8: Receiver Characteristics – BLE

Parameter	Conditions	Min	Typ	Max	Unit
Sensitivity @30.8% PER	-	-	-97	-	dBm
Maximum received signal @30.8% PER	-	0	-	-	dBm
Co-channel C/I	-	-	+10	-	dB
Adjacent channel selectivity C/I	$F = F_0 + 1 \text{ MHz}$	-	-5	-	dB
	$F = F_0 - 1 \text{ MHz}$	-	-5	-	dB
	$F = F_0 + 2 \text{ MHz}$	-	-25	-	dB
	$F = F_0 - 2 \text{ MHz}$	-	-35	-	dB
	$F = F_0 + 3 \text{ MHz}$	-	-25	-	dB
	$F = F_0 - 3 \text{ MHz}$	-	-45	-	dB
Out-of-band blocking performance	30 MHz ~ 2000 MHz	-10	-	-	dBm
	2000 MHz ~ 2400 MHz	-27	-	-	dBm
	2500 MHz ~ 3000 MHz	-27	-	-	dBm
	3000 MHz ~ 12.5 GHz	-10	-	-	dBm
Intermodulation	-	-36	-	-	dBm

5.5.2 Transmitter

Table 9: Transmitter Characteristics – BLE

Parameter	Conditions	Min	Typ	Max	Unit
RF transmit power	-	-	0	-	dBm
Gain control step	-	-	3	-	dBm
RF power control range	-	-12	-	+9	dBm
Adjacent channel transmit power	$F = F_0 \pm 2 \text{ MHz}$	-	-52	-	dBm
	$F = F_0 \pm 3 \text{ MHz}$	-	-58	-	dBm
	$F = F_0 \pm > 3 \text{ MHz}$	-	-60	-	dBm
$\Delta f_{1\text{avg}}$	-	-	-	265	kHz
$\Delta f_{2\text{max}}$	-	247	-	-	kHz
$\Delta f_{2\text{avg}}/\Delta f_{1\text{avg}}$	-	-	-0.92	-	-
ICFT	-	-	-10	-	kHz
Drift rate	-	-	0.7	-	kHz/50 μs
Drift	-	-	2	-	kHz

CD4049



CD4049UB, CD4050B

Data sheet acquired from Harris Semiconductor
SCHS046I

August 1998 - Revised May 2004

CMOS Hex Buffer/Converters

The CD4049UB and CD4050B devices are inverting and non-inverting hex buffers, respectively, and feature logic-level conversion using only one supply voltage (V_{CC}). The input-signal high level (V_{IH}) can exceed the V_{CC} supply voltage when these devices are used for logic-level conversions. These devices are intended for use as CMOS to DTL/TTL converters and can drive directly two DTL/TTL loads. ($V_{CC} = 5V$, $V_{OL} \leq 0.4V$, and $I_{OL} \geq 3.3mA$.)

The CD4049UB and CD4050B are designated as replacements for CD4009UB and CD4010B, respectively. Because the CD4049UB and CD4050B require only one power supply, they are preferred over the CD4009UB and CD4010B in all inverter, current driver, or logic-level conversion applications. In these applications the CD4049UB and CD4050B are pin compatible with the CD4009UB and CD4010B respectively, and can be substituted for these devices in existing as well as in new designs. Terminal No. 16 is not connected internally on the CD4049UB or CD4050B, therefore, connection to this terminal is of no consequence to circuit operation. For applications not requiring high sink-current or voltage conversion, the CD4069UB Hex Inverter is recommended.

Features

- CD4049UB Inverting
- CD4050B Non-Inverting
- High Sink Current for Driving 2 TTL Loads
- High-To-Low Level Logic Conversion
- 100% Tested for Quiescent Current at 20V
- Maximum Input Current of 1 μ A at 18V Over Full Package Temperature Range; 100nA at 18V and 25°C
- 5V, 10V and 15V Parametric Ratings

Applications

- CMOS to DTL/TTL Hex Converter
- CMOS Current "Sink" or "Source" Driver
- CMOS High-To-Low Logic Level Converter

Ordering Information

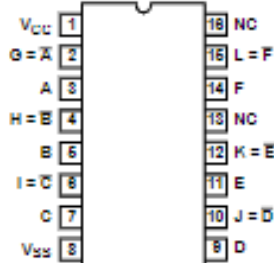
PART NUMBER	TEMP. RANGE (°C)	PACKAGE
CD4049UBF3A	-55 to 125	16 Ld CERDIP
CD4050BF3A	-55 to 125	16 Ld CERDIP
CD4049UBD	-55 to 125	16 Ld SOIC
CD4049UBDR	-55 to 125	16 Ld SOIC
CD4049UBDT	-55 to 125	16 Ld SOIC
CD4049UBDW	-55 to 125	16 Ld SOIC
CD4049UBDWR	-55 to 125	16 Ld SOIC
CD4049UBE	-55 to 125	16 Ld PDIP
CD4049UBNSR	-55 to 125	16 Ld SOP
CD4049UBPW	-55 to 125	16 Ld TSSOP
CD4049UBPWR	-55 to 125	16 Ld TSSOP
CD4050BD	-55 to 125	16 Ld SOIC
CD4050BDR	-55 to 125	16 Ld SOIC
CD4050BDT	-55 to 125	16 Ld SOIC
CD4050BDW	-55 to 125	16 Ld SOIC
CD4050BDWR	-55 to 125	16 Ld SOIC
CD4050BE	-55 to 125	16 Ld PDIP
CD4050NSR	-55 to 125	16 Ld SOP
CD4050BPW	-55 to 125	16 Ld TSSOP
CD4050BPWR	-55 to 125	16 Ld TSSOP

NOTE: When ordering, use the entire part number. The suffix R denotes tape and reel. The suffix T denotes a small-quantity reel of 250.

Pinouts

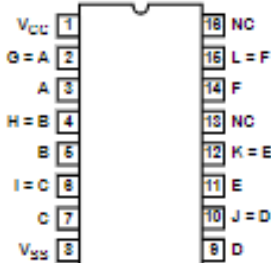
CD4049UB (PDIP, CERDIP, SOIC, SOP, TSSOP)

TOP VIEW



CD4050B (PDIP, CERDIP, SOIC, SOP)

TOP VIEW



CD4049UB, CD4050B**Absolute Maximum Ratings**

Supply Voltage (V+ to V-): -0.5V to 20V
 DC Input Current, Any One Input: ± 10 mA

Operating Conditions

Temperature Range: -55°C to 125°C

Thermal Information

Package Thermal Impedance, θ_{JA} (see Note1):
 E (PDIP) Package 67°C/W
 D (SOIC) Package 73°C/W
 DW (SOIC) Package 57°C/W
 NS (SOP) Package 64°C/W
 PW (TSSOP) Package 108°C/W
 Maximum Junction Temperature (Plastic Package) 150°C
 Maximum Storage Temperature Range -65°C to 150°C
 Maximum Lead Temperature (Soldering 10s) 265°C
 SOIC - Lead Tips Only

CAUTION: Stresses above those listed in "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

NOTE:

1. The package thermal impedance is calculated in accordance with JESD 51-7.

DC Electrical Specifications

PARAMETER	TEST CONDITIONS			LIMITS AT INDICATED TEMPERATURE (°C)							UNITS
				-55	-40	85	125	25			
	V _O (V)	V _{IN} (V)	V _{CC} (V)					MIN	TYP	MAX	
Quiescent Device Current I _{DD} (Max)	-	0,5	5	1	1	30	30	-	0.02	1	µA
	-	0,10	10	2	2	60	60	-	0.02	2	µA
	-	0,15	15	4	4	120	120	-	0.02	4	µA
	-	0,20	20	20	20	600	600	-	0.04	20	µA
Output Low (Sink) Current I _{OL} (Min)	0.4	0,5	4.5	3.3	3.1	2.1	1.8	2.6	5.2	-	mA
	0.4	0,5	5	4	3.8	2.9	2.4	3.2	6.4	-	mA
	0.5	0,10	10	10	9.6	6.6	5.6	8	16	-	mA
	1.5	0,15	15	26	25	20	18	24	48	-	mA
Output High (Source) Current I _{OH} (Min)	4.6	0,5	5	-0.81	-0.73	-0.58	-0.48	-0.65	-1.2	-	mA
	2.5	0,5	5	-2.6	-2.4	-1.9	-1.55	-2.1	-3.9	-	mA
	9.5	0,10	10	-2.0	-1.8	-1.35	-1.18	-1.65	-3.0	-	mA
	13.5	0,15	15	-5.2	-4.8	-3.5	-3.1	-4.3	-8.0	-	mA
Out Voltage Low Level V _{OL} (Max)	-	0,5	5	0.05	0.05	0.05	0.05	-	0	0.05	V
	-	0,10	10	0.05	0.05	0.05	0.05	-	0	0.05	V
	-	0,15	15	0.05	0.05	0.05	0.05	-	0	0.05	V
Output Voltage High Level V _{OH} (Min)	-	0,5	5	4.95	4.95	4.95	4.95	4.95	5	-	V
	-	0,10	10	9.95	9.95	9.95	9.95	9.95	10	-	V
	-	0,15	15	14.95	14.95	14.95	14.95	14.95	15	-	V
Input Low Voltage, V _{IL} (Max) CD4049UB	4.5	-	5	1	1	1	1	-	-	1	V
	9	-	10	2	2	2	2	-	-	2	V
	13.5	-	15	2.5	2.5	2.5	2.5	-	-	2.5	V
Input Low Voltage, V _{IL} (Max) CD4050B	0.5	-	5	1.5	1.5	1.5	1.5	-	-	1.5	V
	1	-	10	3	3	3	3	-	-	3	V
	1.5	-	15	4	4	4	4	-	-	4	V

CD4049UB, CD4050B**DC Electrical Specifications** (Continued)

PARAMETER	TEST CONDITIONS			LIMITS AT INDICATED TEMPERATURE (°C)							UNITS
				-55	-40	85	125	25			
	V _O (V)	V _{IN} (V)	V _{CC} (V)					MIN	TYP	MAX	
Input High Voltage, V _{IH} Min CD4049UB	0.5	-	5	4	4	4	4	4	-	-	V
	1	-	10	8	8	8	8	8	-	-	V
	1.5	-	15	12.5	12.5	12.5	12.5	12.5	-	-	V
Input High Voltage, V _{IH} Min CD4050B	4.5	-	5	3.5	3.5	3.5	3.5	3.5	-	-	V
	9	-	10	7	7	7	7	7	-	-	V
	13.5	-	15	11	11	11	11	11	-	-	V
Input Current, I _{IN} Max	-	0,18	18	±0.1	±0.1	±1	±1	-	±10 ⁻⁵	±0.1	μA

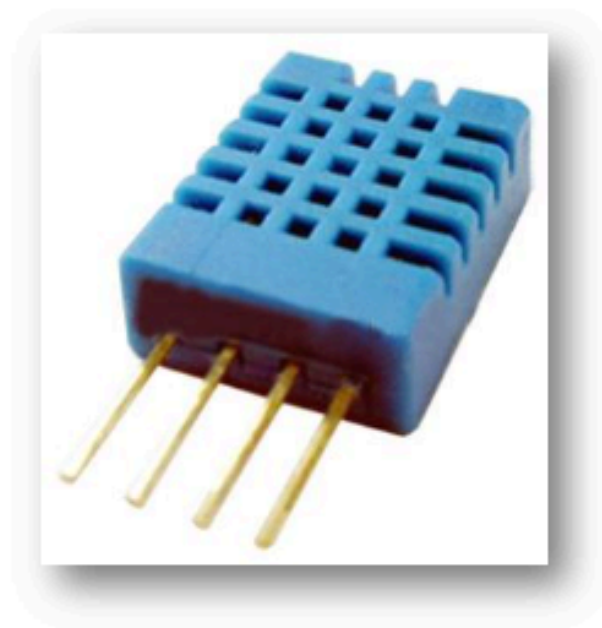
AC Electrical Specifications T_A = 25°C, Input t_r, t_f = 20ns, C_L = 50pF, R_L = 200kΩ

PARAMETER	TEST CONDITIONS		LIMITS (ALL PACKAGES)		UNITS
	V _{IN}	V _{CC}	TYP	MAX	
Propagation Delay Time Low to High, t _{PLH} CD4049UB	5	5	60	120	ns
	10	10	32	65	ns
	10	5	45	90	ns
	15	15	25	50	ns
	15	5	45	90	ns
Propagation Delay Time Low to High, t _{PLH} CD4050B	5	5	70	140	ns
	10	10	40	80	ns
	10	5	45	90	ns
	15	15	30	60	ns
	15	5	40	80	ns
Propagation Delay Time High to Low, t _{PHL} CD4049UB	5	5	32	65	ns
	10	10	20	40	ns
	10	5	15	30	ns
	15	15	15	30	ns
	15	5	10	20	ns
Propagation Delay Time High to Low, t _{PHL} CD4050B	5	5	55	110	ns
	10	10	22	55	ns
	10	5	50	100	ns
	15	15	15	30	ns
	15	5	50	100	ns
Transition Time, Low to High, t _{TLH}	5	5	80	160	ns
	10	10	40	80	ns
	15	15	30	60	ns
Transition Time, High to Low, t _{THL}	5	5	30	60	ns
	10	10	20	40	ns
	15	15	15	30	ns

AC Electrical Specifications T_A = 25°C, Input t_r, t_f = 20ns, C_L = 50pF, R_L = 200kΩ (Continued)

PARAMETER	TEST CONDITIONS		LIMITS (ALL PACKAGES)		UNITS
	V _{IN}	V _{CC}	TYP	MAX	
Input Capacitance, C _{IN} CD4049UB	-	-	15	22.5	pF
Input Capacitance, C _{IN} CD4050B	-	-	5	7.5	pF

DHT11



Each DHT11 element is strictly calibrated in the laboratory that is extremely accurate on humidity calibration. The calibration coefficients are stored as programmes in the OTP memory, which are used by the sensor's internal signal detecting process. The single-wire serial interface makes system integration quick and easy. Its small size, low power consumption and up-to-20 meter signal transmission making it the best choice for various applications, including those most demanding ones. The component is 4-pin single row pin package. It is convenient to connect and special packages can be provided according to users' request.

2. Technical Specifications:

Overview:

Item	Measurement Range	Humidity Accuracy	Temperature Accuracy	Resolution	Package
DHT11	20-90%RH 0-50 °C	± 5 %RH	± 2 °C	1	4 Pin Single Row

Detailed Specifications:

Parameters	Conditions	Minimum	Typical	Maximum
Humidity				
Resolution		1%RH	1%RH	1%RH
			8 Bit	
Repeatability			± 1%RH	
Accuracy	25°C		± 4%RH	
	0-50°C			± 5%RH
Interchangeability	Fully Interchangeable			
Measurement Range	0°C	30%RH		90%RH
	25°C	20%RH		90%RH
	50°C	20%RH		80%RH
Response Time (Seconds)	1/e(63%)25°C, 1m/s Air	6 S	10 S	15 S
Hysteresis			± 1%RH	
Long-Term Stability	Typical		± 1%RH/year	
Temperature				
Resolution		1°C	1°C	1°C
		8 Bit	8 Bit	8 Bit
Repeatability			± 1°C	
Accuracy		± 1°C		± 2°C
Measurement Range		0°C		50°C
Response Time (Seconds)	1/e(63%)	6 S		30 S

6. Electrical Characteristics

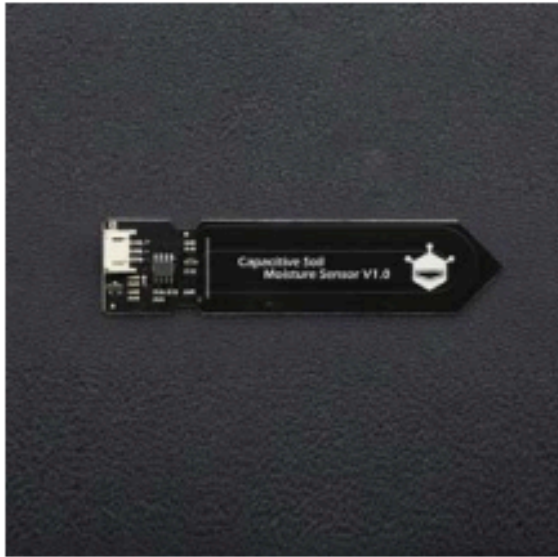
VDD=5V, T = 25°C (unless otherwise stated)

	Conditions	Minimum	Typical	Maximum
Power Supply	DC	3V	5V	5.5V
Current Supply	Measuring	0.5mA		2.5mA
	Average	0.2mA		1mA
	Standby	100uA		150uA
Sampling period	Second	1		

Note: Sampling period at intervals should be no less than 1 second.

Sensor de Humedad del Suelo

Capacitive Soil Moisture Sensor SKU:SEN0193



Capacitive Soil Moisture Sensor

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Introduction

Our soil moisture sensor measures soil moisture levels by capacitive sensing rather than resistive sensing like other sensors on the market. It is made of corrosion resistant material which gives it an excellent service life.

Insert it in to the soil around your plants and impress your friends with real-time soil moisture data! This module includes an on-board voltage regulator which gives it an operating voltage range of 3.3 ~ 5.5V. It is perfect for low-voltage MCUs, both 3.3V and 5V. For compatibility with a Raspberry Pi it will need an ADC converter.

This sensor is compatible with our 3-pin "Gravity" interface, which can be directly connected to the Gravity I/O expansion shield.

Specification

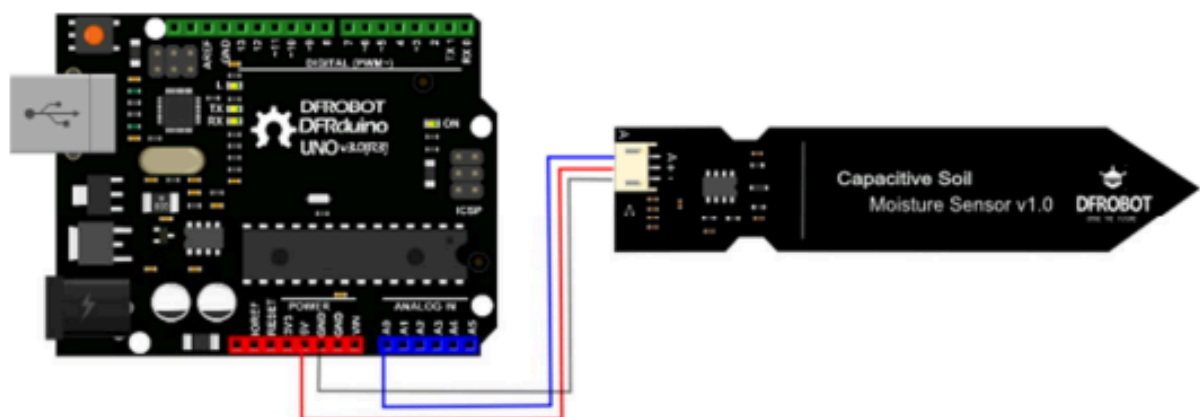
- Operating Voltage: 3.3 ~ 5.5 VDC
- Output Voltage: 0 ~ 3.0VDC
- Operating Current: 5mA
- Interface: PH2.0-3P
- Dimensions: 3.86 x 0.905 inches (L x W)
- Weight: 15g

Tutorial

Requirements

- **Hardware**
UNO x1
Capacitive Soil Moisture Sensor x1
Jumper Cable x3
- **Software**
Arduino IDE V1.6.5 [Click to Download Arduino IDE](#)

Connection Diagram



Calibration Code

```
void setup() {  
  Serial.begin(9600); // open serial port, set the baud rate as 9600 bps  
}  
void loop() {  
  int val;  
  val = analogRead(0); //connect sensor to Analog 0  
  Serial.print(val); //print the value to serial port  
  delay(100);  
}
```

Calibration

Calibration Range

